Supplementary information
Cross-fitted selective recovery under controlled multi-sensor corruption

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1 Algorithm and implementation details

For every training minibatch, PDRF encodes each available sensor group, computes group logits and bounded scores, and forms normalized weights. One available group is sampled from the training prior. The group receives Gaussian degradation whose scale is determined by a uniformly sampled retention factor. The model then evaluates the degraded view and, when another group remains, the selected-group removal view. The main-text objective combines clean classification, bounded score, boundary, degraded classification, consistency, normalized-weight monotonicity and severity-linked score ranking. All random choices use explicitly recorded seeds.

The RO-PDRF family uses the same architecture and data views. In the formal protocol, the detached clean-view probability vector is the teacher for degraded-view KL distillation; it therefore requires no test label or oracle view selection. RO-PDRF-Lite adds only this recovery term at weight 0.30 to the base PDRF objective and is the primary practical model and software default. RO-PDRF-Full additionally applies clean/degraded multiclass Brier terms, a log interior penalty on both response paths and a smooth one-versus-rest pairwise loss on degraded logits at weights 0.05, 0.005 and 0.10; it is the mechanistic full model. For a minibatch $\mathcal{B}$, both task losses use the exact class-weighted mean $\sum_{i\in\mathcal{B}} \omega_{y_i}[-\log p_{iy_i}]/\sum_{i\in\mathcal{B}} \omega_{y_i}$, evaluated on clean and degraded logits respectively. Consistency, recovery and removal-dependent directional terms are averaged only over observations with at least two available groups: removal of the degraded group is otherwise undefined because no predictor remains. With $\epsilon = 10^{-4}$, the interior term is the sum of the clean and degraded available-group means of $-\log\{\max[\epsilon, 1-(r/B)^2]\}$. The pairwise ranking term averages only over classes having at least one positive and one negative minibatch observation and is zero when no class is eligible. RO-PDRF-EMA is a calibration-oriented sensitivity based on Full. Label-assisted best-view, affected-group-removal and confidence-selected rules are separate sensitivity experiments. Development ablations used seeds 101–103; the selected combination was frozen before formal evaluation on seeds 101–110.

Table S1: Eligibility of chemical training observations for the removal-dependent recovery loss. Values summarize 10 fixed training-mask realizations.

Training rows	Recovery-loss eligible	Excluded	Excluded fraction
5933	5910.1 ± 3.1	22.9 ± 3.1	0.004 ± 0.001

The chemical-array group encoder is Linear(32,64, bias=true), ReLU, Linear(64,32, bias=true), ReLU. The hydraulic encoder is Linear(48,64, bias=true), ReLU, Linear(64,32, bias=true), ReLU. The AHU group encoder uses the same architecture with four inputs. The logit head is Linear(32,C, bias=true). The score head is Linear(32,16, bias=true), ReLU, Linear(16,1, bias=true), with no output activation before $3\tanh(u/3)$. The CAGF/DWR gate is Linear($32M + 2M$,64), ReLU, Linear(64,M). Early fusion is Linear($Md + 2M$,128), ReLU, Linear(128,64), ReLU, Linear(64,C). No batch normalization, layer normalization or dropout is used.

Table S2: Empirical bounded-response audit. The reported 3.000000 faulted maximum is a stored float32 value after rounding; the real-arithmetic tanh transformation remains inside the open interval.

View	Minimum	Maximum	Seed–observation responses
Clean	−2.757962	2.997449	17,650 total across views
Faulted	−2.899454	3.000000	10 fitted pairs
Theoretical (real arithmetic)	−3 open	+3 open	–

RO-MER contains the common group classifiers, one joint expert Linear($32M + 2M$,64), ReLU, Linear(64,C), and an independent router Linear($32M+2M$,64), ReLU, Linear(64,$M+1$). The router mixes the M group experts and joint expert after unavailable group experts have been masked. It contains 35,811 parameters for four chemical groups and receives the complete recovery objective. The joint-expert router probability is distributed uniformly over available physical groups only when calculating the shared directional-weight diagnostic; the prediction uses the unmodified $M+1$ expert distribution.

Table S3: Separate architecture-control estimand. Strict fault-applied-and-available macro-AUROC values are means $\pm$ standard deviations across 10 matched optimization seeds. RO-MER includes group experts and a joint expert; all four mechanisms remain controlled interventions on one instrument.

Fault	RO-CAGF	RO-MER	RO-PDRF-Full
gaussian	0.795 ± 0.023	0.787 ± 0.012	0.799 ± 0.010
offset	0.792 ± 0.029	0.807 ± 0.011	0.822 ± 0.013
drift	0.799 ± 0.025	0.813 ± 0.014	0.829 ± 0.013
stuck-at	0.780 ± 0.038	0.797 ± 0.016	0.815 ± 0.011

RO-AT-GATE uses one 32-dimensional token per group, four attention heads (head dimension 8), a 64-unit feed-forward block and dropout 0. Group-identity embeddings are initialized independently from $\mathcal{N}(0, 0.02^2)$. Its shared gate head is Linear(34,16), ReLU, Linear(16,1); the two additional inputs are availability and optional quality. The unweighted selection cross-entropy, 10^{-4} improvement threshold and nine-epoch patience rule are identical to the other models.

Quality-aware Multimodal Fusion (QMF) was originally reported for text–image classification and RGB–D scene recognition. QMF-PD retains the common chemical sensor-group encoder and class heads, computes the signed energy-derived quantity $\bar{c}_m = \mathrm{logsumexp}(\boldsymbol{\mu}_m)/10$, floors it to the positive confidence $c_m = \max(10^{-6}, \bar{c}_m)$, detaches confidence in the unnormalized sum $\sum_m a_m c_m \boldsymbol{\mu}_m$, and assigns unit weight to the historical-loss ranking objective. The floor is required because the unshifted energy-derived quantity can be negative. The pairwise margin is formed from normalized cumulative per-sample cross-entropy histories, matching the published implementation rule. Paired degradation, optimizer, split and checkpoint selection remain common to the present experiment. This is the primary official-rule-aligned sensor-group adaptation; it is not an exact reproduction of the original data or backbones. Sensitivity variants change the confidence scale, fusion normalization, ranking weight and cumulative-history rule one at a time.

All models use AdamW, learning rate 2×10^{-3}, weight decay 10^{-4}, batch size 256 and gradient-norm clipping at 5. Chemical-array models train for at most 60 epochs, hydraulic models for at most 80 and AHU models for at most 12. Chemical checkpoint selection and temperature calibration use disjoint fault-free portions of batch 7 with independently generated 10% group missingness. Checkpoint selection stops after nine epochs without an unweighted cross-entropy improvement above 10^{-4}. Temperature scaling is fitted separately for every seed and method on all observations in its calibration subset by minimizing unweighted multiclass NLL; the same scalar is applied to clean, affected-assignment and unaffected test observations. The formal CEE deployment streams, including the no-fault stream, all use the same fixed 20% group-missingness masks; the earlier natural-view analysis without imposed test missingness is archived separately. Training uses Gaussian paired degradation only; offset, drift and stuck-at are test-time interventions. An observation with no available group is rejected before model inference; the data generator prevents this state during evaluation.

2 Q1 selector-stability, utility and probability audit

The Q1 revision adds frozen run CEE-CF10-R2-LITE. It reuses the chemical split, fixed test realization, optimization seeds 101–110 and training-only preprocessing of CEE-CF10-R2, but routes between PDRF and RO-PDRF-Lite. Six confidence/entropy features are calculated from temperature-scaled endpoint probabilities. The router emits one selected endpoint vector rather than a probability mixture, and no post-routing recalibration is performed. The primary selector uses L_2-regularized logistic regression with $C = 0.05$ and requires two endpoint passes. Its feature scaler is refitted on the informative training rows inside every cross-fitting fold and then refitted on all informative calibration rows for the final model.

Repeated grouped cross-fitting uses 10 split seeds for each of 10 fitted endpoint pairs. All corruption realizations

Table S4: Parameter count and measured CPU cost on the chemical-array run. Training times include early stopping and therefore vary with selected epoch. Inference excludes data loading.

Method	Parameters	Training (s)	Inference (ms/observation)
EF	26,182	1.82	0.000358
EF_PD	26,182	3.36	0.000423
UF	17,560	3.11	0.000507
UF_PD	17,560	4.96	0.000652
Q_ONLY	17,560	4.72	0.000680
CAGF	26,588	7.38	0.000811
DWR	26,588	6.94	0.000855
PDRF_NOQ	19,740	8.42	0.000790
PDRF	19,740	8.74	0.000772

derived from one calibration observation remain in one fold. Across the resulting 100 fits, conditional preference AUROC was 0.720 ± 0.138 and ranged from 0.327 to 0.954. Thresholds ranged from 0.525 to 0.90. The mean primary-objective tie count was 1.78 and the maximum was 9. The formal once-per-pair fit had 1.5 tied thresholds on average (range 1–4). Ties are broken by the highest threshold after realized harms, selected accuracy and retained corrections are equal.

The decision analysis computes $U = V_C N_C - C_H N_H - \lambda(P - 1)$ per 10,000 observations. Harm-to-correction ratios are 1, 2, 5 and 10; pass penalties are 0, 0.5 and 2 correction equivalents per 10,000. The complete source file preserves seed- and mechanism-level values. Energy was not measured. Consequently, neither the pass penalty nor counted FLOPs are presented as electrical-energy estimates.

The all-row three-outcome comparator uses the same six features and calibration rows, including endpoint-equivalent cases. A multinomial logistic model predicts base better, equivalent or recovery better; its routing score is $p(\text{recovery better}) - p(\text{base better})$. The shallow-tree control uses depth 2 and a minimum leaf size of 35. The staged Full-endpoint cascade computes the 12-feature leave-one-group-out selector only for the 25% of out-of-fold six-feature scores nearest the cheap threshold. All thresholds use the same grouped cross-fitting and lexicographic hard-decision objective.

3 Frozen CEE cross-fitted deployment analysis

The formal deployment rerun is identified as CEE-CF10-R2. It uses optimization seeds 101–110 and reuses one PDRF/RO-PDRF-Full checkpoint pair per seed across every prevalence, selector and fault audit. The strict estimand contains an observation only when the controlled intervention reaches an available affected group. For the 40% group-1 assignment, 1,436 of 1,765 assigned observations satisfy this definition and 329 are assigned but masked. The latter cannot transmit the intervention and are reported only as a sensitivity subset.

The second chronological half of the Batch 7 calibration partition contains two method-shared realizations of Gaussian, offset, drift and stuck-at corruption. Only applied-and-available rows are retained. Selector features are base and recovery confidence, their difference, both normalized entropies and their difference, base–recovery Jensen–Shannon (JS) divergence, each endpoint's JS divergence from leave-one-group-out (LOO) recovery consensus, each endpoint's class agreement with that consensus and the legal-removal fraction. The selector contains no class label or fault-type input at inference.

Five-fold stratified group cross-fitting assigns all fault realizations derived from the same original calibration observation to one fold. Within each fold, standardization and logistic coefficients are fitted only on informative rows from the other four folds, where the endpoint models differ in correctness. Pooled out-of-fold conditional preference scores, rather than fitted scores, determine thresholds on the grid $0.50, 0.525, \ldots, 0.90$. The score is conditional on endpoint correctness disagreement and is not a general reliability probability. Every fit uses standardized inputs, L_2 regularization with $C = 0.5$, the `lbfgs` solver, balanced class weights, tolerance 10^{-4}, 2,000 maximum iterations and random state 0. Safe-CF first minimizes negative transfer, then maximizes selector accuracy and retained recovery. Balanced-CF first maximizes accuracy, then minimizes negative transfer. Once thresholds are selected, coefficients are refitted on all informative calibration rows; the thresholds remain frozen. Every training fold contained both

Table S5: Result-to-run manifest for the Q1 revision. Every listed file is stored under `source_data/`; producing scripts and fixed configuration files accompany the submission.

Claim family	Frozen run	Machine-readable evidence
Primary Lite-CF routing	CEE-CF10-R2-LITE	`q1_lite_routing_metrics.csv`, `q1_lite_routing_safety.csv`
Mechanism/batch intervals	CEE-CF10-R2-LITE	`q1_lite_mechanism_fixed_intervals.csv`, `q1_lite_batch_fixed_intervals.csv`
Selector stability	CEE-CF10-R2-LITE	`q1_lite_repeated_grouped_cv.csv`, `q1_lite_regularization_path.csv`, `q1_lite_selector_calibration_size.csv`
Final probability quality	CEE-CF10-R2-LITE	`q1_lite_agreement_probability_quality.csv`, `q1_lite_classwise_metrics.csv`, `q1_lite_probability_by_seed.csv`
Prevalence/cost utility	CEE-CF10-R2-LITE	`q1_lite_prevalence_pass_utility.csv`, `q1_lite_utility_sensitivity_summary.csv`
Direct CPU timing	CEE-CF10-R2-LITE	`q1_lite_cpu_latency.csv`, `q1_lite_cpu_latency_summary.csv`
Full learned baselines/cascade	CEE-CF10-R2-Q1S	`q1_learned_routing_comparison.csv`, `q1_decision_utility.csv`
Response geometry	CEE-CF10-R2-Q1S	`q1_response_range_audit.csv`
Fault plausibility/counts	CEE-CF10-R2-Q1S	`q1_fault_plausibility.csv`, `q1_complete_and_strict_class_counts.csv`

classes, no convergence warning occurred, and test labels are loaded only for evaluation after this process.

The legacy selector audit used 1,000 crossed-cluster bootstrap replicates. Fault mechanism and fitted model-pair seed were sampled with replacement, followed by observation resampling within fault–seed cells. This procedure is preserved for auditability, but four top-level mechanisms are insufficient for reliable mechanism-population resampling. The Q1 revision therefore replaces it in the main article with mechanism-fixed and mechanism–batch-fixed paired intervals. Neither analysis treats 57,440 repeated predictions as independent physical events or represents a population of devices.

The leave-one-family-out audit refits standardization, logistic coefficients and threshold using three calibration families and applies them to the fourth. The unseen audit retains the four-family selector unchanged and applies gain loss, clipping and correlated dual-group corruption, none of which appears in selector calibration. Both audits use the same endpoint checkpoints and fixed test realization as the main run.

Scalar confidence, entropy, LOO-disagreement and JS rules fit thresholds on out-of-fold calibration scores. Prevention-matched rules minimize absolute difference from Safe-CF calibration prevention; the random rule matches recovery-model usage. Always-base, always-recovery and a correctness-disagreement oracle define endpoints. Because the oracle chooses endpoint predictions by hard-decision correctness, it defines an opportunity boundary in prevention–retention space but is not an AUROC upper bound.

Safe-CF performs one PDRF pass, one ordinary RO-PDRF-Full pass and one leave-one-group-out RO-PDRF-Full pass per physical group. Its $G + 2$ forward passes are included in all latency and FLOP statements. CPU measurements use three warm-up calls and seven repeated calls for both the complete 4,364-observation batch and batch size one, excluding data loading. The runtime is an Intel Core i5-13500H with PyTorch 2.12.1+cpu and 12 intra-op and inter-op threads.

The earlier assignment-defined comparisons, teacher heuristics and alternative temperature protocols are retained in the archived sections below for auditability. They are not pooled with CEE-CF10-R2 and do not replace the strict, full-stream or transfer estimands.

4 Fault-subset, batch and class results

Standardization means and standard deviations are fitted on batches 1–6 only. Batches 7–10 are transformed with those frozen quantities and clipped featurewise to $[-8, 8]$ before grouping. Controlled faults are then imposed in standardized grouped space. The formal test fixes all quality values to one. Independent fault assignment and missingness are used to isolate a factorial stress condition; they do not assert a natural joint failure mechanism.

Table S6: Endpoint and final routed-output quality on strict, complete 40%-prevalence and no-imposed-fault streams. Values are equal-weight means over the relevant fixed mechanism–seed cells. ECE uses 15 equal-width bins.

Stream	Method	Accuracy	AUROC	AUPRC	NLL	ECE
Strict	PDRF	0.4869	0.8115	0.5534	2.8239	0.3492
Strict	RO-PDRF-Lite	0.5002	0.8175	0.5643	2.6023	0.3302
Strict	Lite-CF	0.4885	0.8151	0.5598	2.6775	0.3408
Mixed 40%	PDRF	0.5606	0.8426	0.6005	2.2131	0.2696
Mixed 40%	RO-PDRF-Lite	0.5642	0.8451	0.6062	2.1135	0.2628
Mixed 40%	Lite-CF	0.5607	0.8441	0.6043	2.1386	0.2655
No imposed fault	PDRF	0.5907	0.8631	0.6380	1.9002	0.2322
No imposed fault	RO-PDRF-Lite	0.5903	0.8623	0.6378	1.8626	0.2299
No imposed fault	Lite-CF	0.5901	0.8629	0.6385	1.8636	0.2301

Table S7: Probability quality split by endpoint-class agreement on the strict subset. This audit tests the selected-sample concern directly: agreement rows are not automatically harmless in probability space.

Method	Endpoint classes	AUROC	AUPRC	NLL	ECE
PDRF	Agree	0.8177	0.5611	2.8206	0.3488
PDRF	Disagree	0.6776	0.3857	2.8689	0.3759
RO-PDRF-Lite	Agree	0.8235	0.5718	2.6010	0.3419
RO-PDRF-Lite	Disagree	0.7030	0.4176	2.6159	0.2073
Lite-CF	Agree	0.8214	0.5679	2.6670	0.3418
Lite-CF	Disagree	0.6831	0.3908	2.8129	0.3440

The assignment-defined mixed test set combines 40% assigned and 60% unassigned observations. The following archived table separates those subsets. The assignment index is generated once and shared by all methods and seeds; the main text uses the stricter fault-applied-and-available subset.

Seed-level probabilities, predicted classes, true classes, acquisition batches and affected indicators are provided in `source_data/major_seed_predictions.csv`. They permit reconstruction of every confusion matrix and disagreement analysis without rounded table values.

5 Diagnostic-quality controls

No-q-update and missing-q conditions both set $q = 1$. Reported metadata assign $q = 0.15$ to the affected group. Shuffled metadata permute those values over observations. Misleading metadata assign low quality to unaffected groups while keeping the corrupted group at $q = 1$. PDRF-NOQ does not multiply by q during training or inference.

The no-metadata PDRF retains the full method's AUROC, whereas the quality-only rule reaches 0.810. Misleading metadata reduce ordinary PDRF to 0.773. The experiment isolates a benefit from bounded paired training that does not require explicit diagnostics, while showing that using unvalidated metadata creates an avoidable failure route.

6 Sensor grouping and unseen groups

The grouping experiment contains the original consecutive groups, interleaved groups, 10 fixed random four-by-four partitions and a 16-input condition in which every physical sensor is separate. CAGF and PDRF use exactly the same grouping, split, fixed fault and seed within each comparison. The table reports all partitions rather than selecting the best.

Across the 10 random partitions, mean PDRF AUROC ranged from 0.832 to 0.860 and CAGF ranged from 0.812

Table S8: Seed-level dispersion of strict-subset routed-output metrics. Each fitted-pair value first averages the four fixed mechanisms; entries are mean ± standard deviation over 10 fitted endpoint pairs. They quantify fitting variation on the present test observations, not independent devices.

Method	Macro-AUROC	Macro-AUPRC	NLL	Brier	ECE
Lite-CF	0.8151 ± 0.0116	0.5598 ± 0.0194	2.677 ± 0.472	0.813 ± 0.054	0.341 ± 0.041
PDRF	0.8115 ± 0.0115	0.5534 ± 0.0189	2.824 ± 0.474	0.829 ± 0.049	0.349 ± 0.041
RO-PDRF-Lite	0.8175 ± 0.0105	0.5643 ± 0.0173	2.602 ± 0.425	0.799 ± 0.048	0.330 ± 0.042

Table S9: Lite-CF standardized coefficient and sign stability over 500 fold fits. Dominant sign is the larger of positive- and negative-sign frequency; it is not a causal importance measure.

Feature	Median coefficient	IQR	Dominant sign
Base confidence	0.162	[0.068, 0.289]	88.8%
Lite confidence	0.178	[0.124, 0.321]	94.0%
Confidence change	0.061	[0.015, 0.110]	83.0%
Base entropy	−0.180	[−0.264, 0.057]	63.0%
Lite entropy	−0.117	[−0.181, 0.087]	61.2%
Entropy change	0.155	[−0.043, 0.231]	67.2%

to 0.847. PDRF had the higher five-seed mean for every tested grouping. The 16-independent-sensor condition reached 0.813 for PDRF and 0.781 for CAGF. Grouping changes absolute performance and the magnitude of the difference, but did not reverse the mean ordering in this analysis.

The leave-one-group-out experiment assigns zero training degradation probability to each group in turn, then faults that group at test time.

Absolute AUROC declined for both methods. PDRF retained a higher mean than CAGF for all four held-out groups, but the result remains within one 16-sensor device and does not establish new-device generalization.

7 Fault prevalence, severity and location

PDRF and CAGF were evaluated at prevalences 0.10, 0.40 and 0.70; standardized scales 1, 3 and 5; and fault groups 1–4. Each grid point stores affected, unaffected and mixed-set macro-AUROC for 10 matched seeds in `source_data/major_fault_grid.csv`. The principal setting was not selected from this grid. The earlier gain, offset, linear drift, stuck-at, clipping and correlated dual-group tests remain available in the Q2 source files. Training uses one degraded group per pair, whereas the dual-group test evaluates multiple simultaneous faults.

8 Score semantics and directional audit

The raw score increased after degradation in 98.8% of affected observed cases. The affected unnormalized weight decreased in 98.8%, and the normalized weight decreased in 98.4%. The slightly larger normalized violation rate reflects changes in competing groups, but the raw and unnormalized results rule out a purely denominator-driven explanation. Other sensors' normalized weights rose by 0.050 on average.

The synthetic calibration audit fits an isotonic score-to-noise mapping on a separate validation split. Across 20 seeds, test noise MAE was 0.147, RMSE was 0.183 and weighted 10-bin score calibration error was 0.0078 in noise-scale units. These post-hoc values are generator-specific. Raw global Spearman correlation remained 0.403 and cross-sensor pairwise ordering accuracy 0.651. Mean between-sensor raw-score range within matched noise deciles was 0.127. The score therefore admits a useful monotonic calibration within the fixed generator but is not intrinsically aligned across sensors.

9 Hyperparameter sensitivity

The main values $B = 3$, $\beta = 0.25$ and ranking coefficient 0.15 were fixed before the formal rerun. The following three-seed sensitivity analysis is exploratory and is not used for post-hoc model replacement.

Table S10: Lite-CF regularization path. Values average five grouped repeats for each of 10 endpoint pairs. The deployed $C = 0.05$ is a strong-shrinkage choice rather than the calibration-discrimination optimum.

C	OOF AUROC	OOF AUPRC	OOF Brier	Mean threshold
0.01	0.6900	0.7949	0.2255	0.6590
0.05	0.7168	0.8072	0.2083	0.7185
0.10	0.7439	0.8172	0.2010	0.7295
0.50	0.7884	0.8358	0.1854	0.7560
1.00	0.8003	0.8393	0.1792	0.7780
5.00	0.8105	0.8407	0.1735	0.8060

Table S11: Calibration-set-size sensitivity. Values pool seed–repeat summaries and therefore do not share the equal-weight mechanism–seed estimand of the main headline result.

Calibration groups	Prevention	Retention	Recovery use	Macro-AUROC
25%	0.8872	0.1214	0.4837	0.8097
50%	0.9163	0.1143	0.4297	0.8089
75%	0.9242	0.1109	0.4795	0.8102
100%	0.9386	0.1002	0.5076	0.8105

Larger bounds reduced relative boundary occupancy without materially changing AUROC over $B = 3$–5. Larger β increased AUROC in the three-seed grid but also increased saturation. These patterns require an independently selected tuning study before changing the prespecified method.

10 Independent hydraulic single-rig sensitivity analysis

The test-rig repository contains 1,449 cycles marked stable. For each component task, the same fixed proportions are used for training, checkpoint selection, calibration and testing. The class-condition values and counts are listed below.

The raw physical sensors are PS1–PS6 (pressure), FS1–FS2 (flow), TS1–TS4 (temperature), EPS1 (motor power) and VS1 (vibration). Three computed virtual channels are excluded. Every sensor contributes mean, standard deviation, minimum, maximum, first and third quartiles, root-mean-square value and linear slope. Physical-group summary vectors are linearly resampled to 48 inputs. This operation changes dimensional representation but introduces no new observations or labels.

Across the guarded blocked analysis, cooler performance was near ceiling, valve performance was moderate, and pump and accumulator were near chance. The task-averaged affected macro-AUROC values for RO-PDRF-Full were 0.996, 0.756, 0.528 and 0.487, respectively; the corresponding RO-CAGF values were 0.945, 0.735, 0.551 and 0.476. Six of eight task–fault means favoured RO-PDRF-Full, but architecture differences on the near-chance tasks require particular caution. These results do not convert the injected sensor corruptions into native failures; only the component-condition targets are native to the physical test rig.

11 Recovery, calibration and mechanism audits

All methods use a temperature fitted separately for each seed. RO-PDRF-Full improves affected NLL, Brier score and ECE relative to PDRF, but all three remain worse than CAGF. Equal probability averaging yields all-sample AUROC 0.8509 for RO-PDRF-Full and 0.8537 for CAGF, whereas affected-sample AUROC remains 0.0192 lower for Full. The manuscript therefore separates repeated single-model recovery from fixed-ensemble probability quality.

The prediction-level files permit direct auditing of each link claimed by the training design. They include clean and fault correctness, true-class-probability change, bounded-response change, normalized-weight reduction, response saturation, class labels and seed identifiers. Downweighting succeeds on 80.2% of affected RO-PDRF-Full observations, but 41.9% of all affected observations fail despite that decrease. The supplied failure file lists the 40 largest such reductions so that the negative cases are inspectable rather than summarized away.

Table S12: Batch-fixed Lite-CF minus PDRF macro-AUROC. Each cell holds a paired mean and 95% Student-t interval over 10 fitted model pairs while mechanism, batch and observations remain fixed.

Mechanism	Batch 8	Batch 9	Batch 10
Gaussian	0.0062 [0.0019, 0.0106]	0.0077 [0.0036, 0.0117]	0.0061 [0.0032, 0.0089]
Offset	0.0074 [−0.0003, 0.0151]	0.0069 [−0.0012, 0.0150]	0.0019 [−0.0003, 0.0042]
Drift	0.0023 [−0.0004, 0.0051]	0.0032 [−0.0001, 0.0065]	0.0013 [−0.0005, 0.0032]
Stuck-at	0.0031 [−0.0007, 0.0069]	0.0058 [−0.0006, 0.0121]	0.0032 [0.0006, 0.0059]

Table S13: Learned routing and compute controls moved from the main article. Full-endpoint rows route between PDRF and RO-PDRF-Full; Lite-CF routes between PDRF and RO-PDRF-Lite. Prevention and retention are hard-decision outcomes, whereas AUROC and NLL evaluate the emitted probability vector.

Method	Prevention ↑	Retention ↑	Recovery use	Passes	AUROC ↑	NLL ↓
PDRF endpoint	1.000	0.000	0.000	1.00	0.8115	2.8239
RO-PDRF-Full endpoint	0.000	1.000	1.000	1.00	0.8162	2.6244
Cascade-25	**0.959**	0.053	0.473	2.91	0.8149	2.6492
Safe-CF-12	0.900	**0.131**	0.440	6.00	0.8146	2.6927
Three-outcome-6	0.944	0.085	0.046	2.00	0.8117	2.8104
Tree-6	0.919	0.094	0.245	2.00	0.8120	2.7508
Lite-CF (primary)	0.947	0.097	0.522	2.00	**0.8151**	2.6775

Table S43: Paired conditional recovery and error-transfer audit across five fitted RO-PDRF-Full models and eight method-shared fault realizations. The clean-teacher state is taken from the actual Full-model clean forward pass; the baseline state is ordinary PDRF under the matched fault. Rows are repeated predictions from one instrument and are not independent devices.

Clean teacher	Baseline fault prediction	Eligible	Changed	Conditional rate	Outcome
Correct	Wrong	10558	2293	0.217	Recovered by RO-PDRF-Full
Wrong	Correct	4001	1177	0.294	Correction lost under RO-PDRF-Full

Table S44: Modern uncertainty-weighted comparator under the fixed chemical-array fault. ENT-PD uses normalized per-group predictive entropy for positive fusion weights and otherwise receives the same paired-degradation protocol. Values are means across 10 matched seeds.

Method	All AUROC	All AUPRC	All NLL	All ECE	Aff. AUROC	Aff. AUPRC	Aff. NLL	Aff. ECE
CAGF	0.817	0.542	1.828	0.233	0.792	0.505	2.095	0.277
ENT-PD	0.804	0.537	2.124	0.258	0.754	0.467	2.848	0.348
PDRF	0.834	0.580	2.345	0.289	0.791	0.509	3.085	0.369
RO-PDRF-Full	0.840	0.589	2.154	0.277	0.804	0.527	2.693	0.344

RO-PDRF-Full exceeded ENT-PD in affected macro-AUROC for all 10 matched seeds (mean difference 0.0497; exact two-sided Wilcoxon $P = 0.0020$). ENT-PD was also lower than PDRF on the affected subset by 0.0366 across all seeds. The result indicates that predictive confidence and intervention-specific recoverability are not interchangeable.

12 Strong learned baseline and teacher/calibration remedies

RO-AT-GATE retains the four physical sensor groups and the same per-group encoders as the other late-fusion models. Each 32-dimensional group representation receives a learned identity token and passes through one four-head self-attention layer with feed-forward width 64. Unavailable groups are attention-masked. A shared head maps the contextualized group token, availability and quality to a softmax gate. The resulting model has 26,809

Table S14: Frozen run manifest. The test realization, temperature protocol, checkpoint pair and selector protocol are common to all CEE-CF10-R2 headline tables.

Item	Frozen setting
Run identifier	CEE-CF10-R2
Optimization seeds	101–110
Training rerun	major.fit-v2026-07-14-clean-teacher
Test realization	70001
Temperature protocol	First chronological half of Batch 7
Selector protocol	Second half; five-fold sample-group cross-fitting
Checkpoint reuse	One PDRF/Full pair reused across all selector audits per seed

Table S15: Strict fault-applied-and-available macro-AUROC means ± standard deviations across 10 matched optimization seeds. Each mechanism contains 1,436 observations per seed.

Fault mechanism	PDRF	RO-PDRF-Full	Safe-CF	Full–PDRF	Safe–PDRF
Gaussian noise	0.7842 ± 0.0113	0.7986 ± 0.0104	0.7909 ± 0.0107	+0.0143 [0.0103, 0.0183]	+0.0066 [0.0040, 0.0093]
Offset	0.8207 ± 0.0144	0.8216 ± 0.0131	0.8229 ± 0.0114	+0.0009 [−0.0079, 0.0097]	+0.0022 [−0.0013, 0.0058]
Drift	0.8292 ± 0.0141	0.8293 ± 0.0135	0.8307 ± 0.0123	+0.0001 [−0.0078, 0.0081]	+0.0015 [−0.0018, 0.0048]
Stuck-at	0.8118 ± 0.0108	0.8153 ± 0.0107	0.8139 ± 0.0089	+0.0035 [−0.0038, 0.0108]	+0.0021 [−0.0012, 0.0054]

parameters, compared with 26,588 for CAGF, and receives the same clean/faulted views, recovery distillation, Brier term, degraded-AUC term and gate-interior regularizer as RO-CAGF. This is a controlled sensor-group attention baseline, not an exact reproduction of a task-specific image, language or autonomous-driving network.

The EMA teacher follows mean-teacher weight averaging and supplies the clean-view probability target; the student alone is retained for inference. Decays 0.95, 0.99 and 0.995 are evaluated. The hard-threshold control includes only teacher predictions with maximum probability at least 0.70. For continuous weighting, $c_i = \max_c p_{ic}$ and the minibatch recovery term is $\sum_i c_i D_{\mathrm{KL}}(\mathbf{p}_i \| \mathbf{p}_i')/(\sum_i c_i + 10^{-6})$. Clean/removal agreement (CRA) averages clean and affected-group-removal probabilities and activates distillation only when their argmax classes agree. Student/EMA agreement (ECA) applies the same rule to student-clean and EMA-clean probabilities. The CE-conflict training audit activates clean-to-fault distillation during training when clean-view cross-entropy does not exceed degraded-view cross-entropy. It uses labels already present in the two task losses, but the analogous held-out status is not observable without the test label and is not proposed as a deployment selector. The calibration-aware control increases the Brier coefficient from 0.05 to 0.20 without changing any other complete-objective term. All choices were fixed before the revision runs, and every method used the same split, fault realization, minibatch construction, checkpoint rule and calibration procedure.

Table S45: Strong-baseline, teacher and calibration controls under the unique chemical headline fault. Values are means across 10 matched optimization seeds after per-seed temperature scaling. These seeds are repeated fits, not devices.

Method	Params	Aff. AUROC	Aff. AUPRC	Aff. NLL	Aff. Brier	Aff. ECE
RO-AT-GATE	26,809	0.784	0.504	1.826	0.736	0.210
PDRF+CD	19,740	0.802	0.526	2.717	0.809	0.342
PDRF+EMA-CD	19,740	0.803	0.525	2.456	0.800	0.328
RO-PDRF-Full	19,740	0.804	0.525	2.650	0.809	0.342
RO-PDRF-EMA	19,740	0.806	0.527	2.413	0.792	0.321
RO-PDRF-Full-CAL	19,740	0.804	0.528	2.681	0.809	0.342

Table S16: Selector standardization and final logistic coefficients. Values are means ± standard deviations across 10 independently fitted endpoint pairs. Sign agreement is the larger fraction sharing either coefficient sign and is a stability diagnostic, not evidence of causal feature importance. Centers and scales are fitted within each seed; exact seed-level values are provided in the source-data archive.

Feature	Center	Scale	Coefficient	Sign agreement
Base confidence	0.578 ± 0.046	0.145 ± 0.016	0.121 ± 0.256	0.60
Recovery confidence	0.564 ± 0.043	0.147 ± 0.023	0.594 ± 0.307	1.00
Confidence difference	-0.014 ± 0.015	0.173 ± 0.029	0.407 ± 0.264	0.90
Base normalized entropy	0.520 ± 0.050	0.166 ± 0.017	-0.181 ± 0.337	0.80
Recovery normalized entropy	0.541 ± 0.044	0.168 ± 0.023	0.053 ± 0.449	0.50
Entropy difference	0.021 ± 0.011	0.126 ± 0.033	0.251 ± 0.511	0.70
Base–recovery JS divergence	0.080 ± 0.032	0.073 ± 0.019	0.801 ± 0.626	0.80
Base–LOO-consensus JS	0.068 ± 0.030	0.069 ± 0.027	-0.413 ± 0.332	0.90
Recovery–LOO-consensus JS	0.039 ± 0.014	0.046 ± 0.009	0.335 ± 0.456	0.80
Base–LOO class agreement	0.384 ± 0.156	0.462 ± 0.041	0.497 ± 0.294	1.00
Recovery–LOO class agreement	0.551 ± 0.157	0.474 ± 0.024	-0.717 ± 0.427	0.90
Legal-removal fraction	0.844 ± 0.019	0.178 ± 0.016	0.268 ± 0.598	0.50

Table S17: Seed-level selector sample sizes, class counts, frozen thresholds and out-of-fold diagnostics. Calibration n counts applied-and-available rows; disagreement n counts rows informative for logistic fitting. Recovery-positive and negative-transfer columns confirm that both preference classes are present.

Seed	Calibration n	Disagreement n	Recovery positive	Negative transfer	Safe threshold	Balanced threshold	AUROC	AUPRC	Brier
101	1797	177	94	83	0.800	0.675	0.888	0.883	0.135
102	1840	89	40	49	0.900	0.575	0.894	0.879	0.126
103	1860	90	49	41	0.825	0.500	0.852	0.863	0.155
104	1868	89	67	22	0.875	0.550	0.938	0.981	0.093
105	1866	194	122	72	0.800	0.600	0.866	0.929	0.143
106	1869	92	62	30	0.875	0.500	0.949	0.971	0.071
107	1829	224	99	125	0.900	0.500	0.791	0.765	0.188
108	1853	122	54	68	0.900	0.525	0.791	0.735	0.188
109	1880	122	108	14	0.725	0.550	0.839	0.972	0.140
110	1947	99	56	43	0.550	0.550	0.916	0.877	0.095

Table S18: Legacy Full-endpoint cross-fitted selector sensitivity. Out-of-fold discrimination and Brier values summarize 10 fitted pairs. Prevention and retention use the earlier 1,000-replicate crossed bootstrap. With only four top-level mechanisms, these intervals are not used for Q1 headline inference and must not be read as fault-population intervals.

Quantity	Estimate
OOF selector AUROC	0.872 ± 0.056
OOF selector AUPRC	0.886 ± 0.084
OOF selector Brier score	0.133 ± 0.039
Preference events: recovery / negative transfer	75.1 / 54.7 per seed
Minority events per variable	4.15 ± 2.27
Converged final selector fits	10/10; no one-class fold
Safe threshold, median (range)	0.850 (0.550–0.900)
Negative-transfer prevention	0.903 [0.823, 0.952]
Recovery retention	0.144 [0.056, 0.262]
No-fault recovery-selection rate	0.414 ± 0.270
No-fault harmful-switch rate	0.001 ± 0.001

Table S19: Out-of-fold conditional preference-score reliability. Counts pool held-out-fold scores across 10 fitted pairs; the mean score and observed recovery preference are weighted by bin counts. Empty seed–bin combinations contribute no pseudo-counts. The score is not claimed as a generally calibrated reliability probability.

Probability bin	n	Mean conditional preference score	Observed recovery preference
0.0–0.1	135	0.058	0.119
0.1–0.2	149	0.150	0.168
0.2–0.3	134	0.249	0.216
0.3–0.4	96	0.347	0.469
0.4–0.5	74	0.446	0.405
0.5–0.6	81	0.546	0.593
0.6–0.7	87	0.653	0.736
0.7–0.8	123	0.750	0.862
0.8–0.9	135	0.856	0.896
0.9–1.0	284	0.964	0.940

Table S46: Published/adapted baselines and teacher controls under the principal fault. The Role column distinguishes comparison baselines, the practical default, mechanistic full model, calibration sensitivity and diagnostic audits. QMF-PD is the official-rule-aligned sensor-group adaptation; CRA and ECA denote clean/removal agreement and student/EMA agreement. The CE-conflict training audit is label-assisted and training-only. Lite adds clean recovery distillation; Full additionally adds the Brier, ranking and response-interior auxiliaries. Lite, Full, EMA and all teacher controls use the same checkpoint criterion.

Method	Role	Parameters	Seeds	Affected AUROC	NLL	Brier	ECE
QMF-PD	Published-method adaptation	17,560	10	0.775	2.514	0.792	0.287
RO-AT-GATE	Strong learned baseline	26,809	10	0.784	1.826	0.736	0.210
RO-CAGF	Architecture-matched baseline	26,588	10	0.797	2.071	0.770	0.275
RO-PDRF-Lite	Practical default	19,740	10	0.802	2.717	0.809	0.342
RO-PDRF-Full	Mechanistic full model	19,740	10	0.804	2.650	0.809	0.342
RO-PDRF-EMA	Calibration sensitivity	19,740	10	0.806	2.413	0.792	0.321
RO-PDRF-Full-CRA	Diagnostic audit	19,740	10	0.803	2.730	0.819	0.349
RO-PDRF-Full-ECA	Diagnostic audit	19,740	10	0.805	2.494	0.804	0.332
CE-conflict training audit	Diagnostic audit	19,740	10	0.806	2.592	0.801	0.336

Table S20: Complete deployment-stream macro-AUROC, accuracy and macro-F1. Clean/no-fault predictions occur once per seed. Other mechanism-specific values are averaged within seed before the 10 seed units are summarized, preventing unaffected predictions repeated across fault mechanisms from becoming pseudo-replicates.

Evaluation stream	PDRF AUROC	PDRF Acc. / F1	Full AUROC	Full Acc. / F1	Safe-CF AUROC	Safe-CF Acc. / F1
Clean / no imposed fault	0.8631 ± 0.0038	0.591/0.567	0.8634 ± 0.0053	0.591/0.563	0.8632 ± 0.0044	0.590/0.567
Mixed stream, 10% prevalence	0.8587 ± 0.0043	0.584/0.562	0.8597 ± 0.0057	0.584/0.559	0.8591 ± 0.0049	0.584/0.562
Mixed stream, 40% prevalence	0.8426 ± 0.0070	0.561/0.543	0.8450 ± 0.0077	0.562/0.541	0.8439 ± 0.0068	0.561/0.544
Mixed stream, 70% prevalence	0.8288 ± 0.0093	0.534/0.519	0.8328 ± 0.0095	0.537/0.518	0.8312 ± 0.0082	0.535/0.520
Assigned but masked, 40%	0.8446 ± 0.0126	0.570/0.543	0.8448 ± 0.0112	0.566/0.540	0.8453 ± 0.0120	0.569/0.542
Unaffected rows, 40%	0.8667 ± 0.0039	0.600/0.574	0.8669 ± 0.0055	0.600/0.570	0.8668 ± 0.0047	0.600/0.573

Table S21: Operational Safe-CF event counts per 10,000 complete-stream observations. Values are means ± standard deviations across 10 seed-level averages over four mechanisms.

Fault prevalence	Full-model harm prevented / 10,000	Full-model corrections retained / 10,000	Residual Full-model harm / 10,000	Net correct vs PDRF / 10,000
10%	171.5 ± 124.8	11.5 ± 8.9	15.1 ± 12.5	-3.6 ± 15.2
40%	185.6 ± 157.6	20.9 ± 13.4	17.4 ± 15.3	3.4 ± 16.4
70%	192.7 ± 186.5	29.3 ± 13.8	21.5 ± 19.1	7.8 ± 18.5

Table S47: Prediction-level teacher-safeguard audit on the five seeds shared with the original clean-teacher conditional analysis. CRA active coverage includes the valid affected-group-removal requirement. CEG is a training rule and has no label-free test-time coverage. Recovery and transfer rates are conditional paired associations, not causal effects.

Teacher rule or training audit	Active target coverage	Conditional recovery	Conditional error transfer
Clean teacher	100.0%	295/1387 (21.3%)	167/518 (32.2%)
Clean/removal agreement	65.3%	253/1030 (24.6%)	95/289 (32.9%)
Student/EMA agreement	72.4%	356/1265 (28.1%)	156/387 (40.3%)
CE-conflict training audit	Training-only; not observable at test	345/1394 (24.7%)	184/516 (35.7%)

The complete-objective EMA extension changed affected macro-AUROC from 0.8043 to 0.8059 (6/10 positive differences; $P = 0.2324$), NLL from 2.650 to 2.413 (9/10 improvements; $P = 0.0098$), Brier score from 0.809 to 0.792 (8/10; $P = 0.0645$) and ECE from 0.342 to 0.321 (9/10; $P = 0.0059$). A fourfold Brier coefficient did not improve NLL or ECE relative to Full. RO-AT-GATE improved probability quality but reached 0.784 affected AUROC. The official-rule-aligned QMF-PD reached 0.775, NLL 2.514 and ECE 0.287. CEG reached AUROC 0.806 but retained 35.7% conditional error transfer. These results identify EMA smoothing as a probability-quality remedy while preserving the ranking/calibration and teacher-error boundaries.

Table S48: Original QMF implementation choices and their sensor-group adaptation. The primary adaptation aligns confidence scale, detached unnormalized fusion, cumulative history and unit ranking weight with the published implementation; task-specific backbones are not reproduced.

Version	Task/input	Confidence	Fusion	Ranking objective
Original QMF	Text–image; RGB–D	$\mathrm{LSE}(\boldsymbol{\mu})/10$	Unnormalized; confidence detached	Cumulative CE history; normalized margin; weight 1
Primary adaptation	Four chemical sensor groups	$\max(10^{-6}, \bar{c})$	Unnormalized; confidence detached	Cumulative CE history; normalized margin; weight 1
Sensitivity variants	Same groups and paired fault	Scales 0.03, 0.10 or 0.30	Normalized or unnormalized	Weights 0, 0.10 or 1; instantaneous or cumulative history

Table S22: Selector feature-set ablation. Calibration discrimination uses out-of-fold disagreement rows; strict-test quantities first average mechanisms within fitted seed. The all-feature selector was fixed before test evaluation and was not selected using test AUROC.

Feature set	OOF AUROC	Prevention	Retention	Test AUROC
Confidence and entropy (6)	0.782 ± 0.094	0.942 ± 0.064	0.061 ± 0.063	0.8156 ± 0.0096
Disagreement only (5)	0.841 ± 0.087	0.875 ± 0.132	0.101 ± 0.100	0.8119 ± 0.0116
All features (12)	0.872 ± 0.056	0.900 ± 0.113	0.131 ± 0.084	0.8146 ± 0.0094

Table S23: Operating-point sensitivity at 40% fault prevalence. Balanced-CF prioritizes calibration accuracy; Safe-CF prioritizes negative-transfer prevention. Values first average mechanisms within seed.

Operating point	Prevention	Retention	Full use	Accuracy
Balanced-CF	0.758 ± 0.101	0.268 ± 0.108	0.663 ± 0.255	0.562 ± 0.010
Safe-CF	0.895 ± 0.100	0.109 ± 0.075	0.424 ± 0.273	0.561 ± 0.010

Table S49: QMF-specific fairness and hyperparameter sensitivity. Clean results use the clean view with exactly the same availability mask as the faulted view. The primary official-rule-aligned adaptation uses 10 seeds; the controlled variants use five common seeds. Lower NLL and ECE are better.

QMF sensor-group setting	Seeds	Clean AUROC	Affected AUROC	Affected NLL	Affected ECE
Previous adaptation	5	0.840	0.742	3.728	0.391
Confidence scale 0.03	5	0.846	0.762	2.983	0.336
Confidence scale 0.30	5	0.845	0.748	3.530	0.374
Normalized fusion	5	0.844	0.757	3.000	0.367
Ranking weight 0	5	0.840	0.742	3.738	0.391
Ranking weight 1	5	0.845	0.744	3.454	0.375
Official-rule-aligned adaptation	10	0.845	0.775	2.514	0.287

The previous adaptation, normalized fusion and official-rule-aligned implementation all converged without exploding gradients or non-finite values (Fig. S1). Across the five common seeds, changing confidence scale, normalization and ranking produced affected AUROC 0.742–0.768 and ECE 0.296–0.391 while clean AUROC remained 0.840–0.847. Completing the primary adaptation to 10 seeds gave affected AUROC 0.775 ± 0.020, NLL 2.514 ± 0.459 and ECE 0.287 ± 0.046. These results replace the weaker previous QMF point in all main comparisons.

Table S24: Selector transfer to held-out and unseen mechanisms. Prevention and retention are defined relative to endpoint correctness on the strict subset.

Audit	Held-out mechanism	Safe–PDRF	Safe–Full	Prevention	Retention
Leave-one-family-out	Drift	+0.0016	+0.0014	0.912	0.115
Leave-one-family-out	Gaussian noise	+0.0068	-0.0075	0.788	0.324
Leave-one-family-out	Offset	+0.0019	+0.0010	0.825	0.118
Leave-one-family-out	Stuck-at	+0.0024	-0.0011	0.923	0.060
Unseen mechanism	Clipping	-0.0002	-0.0052	0.897	0.049
Unseen mechanism	Correlated Dual	+0.0060	-0.0075	0.744	0.270
Unseen mechanism	Gain Loss	+0.0001	-0.0002	0.963	0.026

Table S25: Matched simple-selector audit across four mechanisms and 10 fitted pairs. Macro-AUROC is mean ± standard deviation over 40 mechanism–seed cells.

Selection rule	Matching	Prevention	Retention	Full use	Macro-AUROC
Safe-CF-12	proposed	0.900	0.131	0.440	0.8146 ± 0.0184
Higher confidence	prevention matched	0.941	0.088	0.034	0.8116 ± 0.0206
Lower entropy	prevention matched	0.933	0.070	0.057	0.8118 ± 0.0205
Lower LOO disagreement	prevention matched	0.919	0.057	0.022	0.8117 ± 0.0205
Higher JS disagreement	prevention matched	0.927	0.076	0.010	0.8116 ± 0.0206
Random matched selection	selection rate matched	0.473	0.491	0.509	0.8123 ± 0.0173
Always PDRF	endpoint	1.000	0.000	0.000	0.8115 ± 0.0210
Always RO-PDRF-Full	endpoint	0.000	1.000	1.000	0.8162 ± 0.0163
Correctness opportunity boundary	opportunity boundary	1.000	1.000	0.029	0.8141 ± 0.0210

Table S26: Direct CPU timing for the primary Lite-CF endpoint chain. Each entry is the mean ± standard deviation of the per-seed median over 10 fitted endpoint pairs. Every median follows three warm-up and seven timed calls; loading is excluded. These workstation measurements are not embedded or real-time guarantees.

Method	Batch 1 latency (ms)	Batch 4,364 latency (ms)	Throughput (obs/s)
Lite-CF	2.963 ± 0.569	12.50 ± 1.72	355644 ± 53332
PDRF	0.734 ± 0.209	4.56 ± 0.59	976286 ± 165789
RO-PDRF-Lite	0.748 ± 0.215	4.48 ± 0.51	986047 ± 122961

Table S27: Four-group deployment cost. State assumes 32-bit parameters. Complete-batch throughput and batch-size-one latency are reported separately; neither is a hardware-independent real-time guarantee.

Method	Parameters	State (KiB)	Passes	FLOPs / sample	Batch throughput / s	Batch-1 latency (ms)
PDRF	19,740	77.1	1	38,528	676997	0.875
RO-PDRF-Full	19,740	77.1	1	38,528	678381	0.880
Safe-CF	39,493	154.3	6	231,168	84288	6.810

Table S28: Counted linear-layer FLOPs and selector forward passes as the number of groups grows.

Groups	PDRF FLOPs	Safe-CF passes	Safe-CF FLOPs
2	19,264	4	77,056
4	38,528	6	231,168
8	77,056	10	770,560
16	154,112	18	2,774,016

Table S29: Chemical-array class counts in the three held-out acquisition batches.

Batch	Class 1	Class 2	Class 3	Class 4	Class 5	Class 6	Total
8	30	30	40	33	143	18	294
9	61	55	100	75	78	101	470
10	600	600	600	600	600	600	3600

Table S30: Strict applied-and-available class counts by held-out acquisition batch. Counts are identical across the four formal mechanisms because assignment and missingness masks are shared.

Batch	Class 1	Class 2	Class 3	Class 4	Class 5	Class 6	Total
8	9	10	10	14	42	6	91
9	28	14	30	32	27	34	165
10	193	196	190	207	197	197	1,180
Total	230	220	230	253	266	237	1,436

Table S31: Severity audit against the empirical training distribution at scales 1, 3 and 5. Percentile shifts are absolute changes in empirical cumulative probability. "Within training range" means the empirical minimum–maximum after training-only standardization and clipping, not a physical sensor limit. The formal scale-3 setting is a severe stress test.

Scale	Mechanism	Median $\lvert\Delta z\rvert$	Median percentile shift	Within 1–99%	Within training range
1	Gaussian	0.672	0.234	0.604	0.745
1	Offset	1.000	0.373	0.561	0.774
1	Drift	0.629	0.253	0.662	0.852
1	Stuck-at	1.137	0.462	0.688	0.875
3	Gaussian	2.015	0.360	0.379	0.609
3	Offset	3.000	0.538	0.332	0.632
3	Drift	1.888	0.440	0.457	0.697
3	Stuck-at	3.105	0.557	0.094	0.656
5	Gaussian	3.358	0.394	0.255	0.532
5	Offset	5.000	0.561	0.005	0.573
5	Drift	3.146	0.516	0.253	0.629
5	Stuck-at	5.097	0.563	0.000	0.563

Table S32: Representative inverse-transformed scale-3 examples from one Batch 8 strict observation. Features are engineered gas-response statistics, not raw voltages; their heterogeneous scales explain the large inverse-transformed magnitudes. The examples demonstrate severity rather than physical plausibility. The complete example file contains five observations per mechanism.

Mechanism	Example feature	Before	After	Before percentile	After percentile
Gaussian	Feature 1	311.875	−38,926.221	0.0062	0.0000
Gaussian	Feature 2	1.196	87.388	0.0133	0.9993
Offset	Feature 1	311.875	−258,238.238	0.0062	0.0000
Offset	Feature 2	1.196	58.822	0.0133	0.9975
Drift	Feature 1	311.875	−64,414.545	0.0062	0.0000
Drift	Feature 2	1.196	15.622	0.0133	0.8952
Stuck-at	Feature 1	311.875	−181,557.895	0.0062	0.0000
Stuck-at	Feature 2	1.196	65.798	0.0133	0.9985

Table S33: Chemical-array affected, unaffected and mixed-set performance under the fixed silent fault. Values are means across 10 seeds.

Method	Subset	Accuracy	Macro-F1	Macro-AUPRC	Macro-AUROC
CAGF	affected	0.474	0.454	0.505	0.792
CAGF	unaffected	0.544	0.518	0.576	0.836
CAGF	all	0.516	0.493	0.542	0.817
PDRF	affected	0.476	0.471	0.509	0.791
PDRF	unaffected	0.600	0.574	0.643	0.867
PDRF	all	0.550	0.535	0.580	0.834

Table S34: Results reported separately for acquisition batches 8, 9 and 10. Values are means across 10 seeds.

Method	Batch	Accuracy	Macro-F1	Macro-AUROC
CAGF	8	0.649	0.596	0.884
CAGF	9	0.581	0.552	0.878
CAGF	10	0.497	0.471	0.808
PDRF	8	0.661	0.623	0.903
PDRF	9	0.588	0.552	0.885
PDRF	10	0.536	0.522	0.824

Table S35: Per-class chemical-array performance under the fixed silent fault. Values are means across 10 seeds. Classes correspond to the six gases in the repository encoding.

Method	Class	Precision	Recall	F1	AUROC	AUPRC
CAGF	1	0.433	0.544	0.475	0.774	0.518
CAGF	2	0.634	0.735	0.676	0.876	0.614
CAGF	3	0.820	0.532	0.631	0.862	0.733
CAGF	4	0.308	0.080	0.124	0.753	0.296
CAGF	5	0.611	0.658	0.632	0.843	0.700
CAGF	6	0.367	0.531	0.421	0.795	0.394
PDRF	1	0.410	0.584	0.477	0.789	0.529
PDRF	2	0.752	0.689	0.717	0.893	0.737
PDRF	3	0.775	0.685	0.726	0.906	0.746
PDRF	4	0.326	0.115	0.169	0.768	0.313
PDRF	5	0.678	0.664	0.670	0.854	0.750
PDRF	6	0.392	0.544	0.450	0.793	0.404

Table S36: Diagnostic-quality controls under the fixed chemical-array fault.

Training	Test metadata	Macro-AUROC	Seed SD
PDRF	misleading	0.773	0.008
PDRF	missing_q	0.834	0.006
PDRF	reported	0.853	0.006
PDRF	shuffled	0.837	0.007
PDRF	silent	0.834	0.006
PDRF_NOQ	misleading	0.835	0.007
PDRF_NOQ	missing_q	0.835	0.007
PDRF_NOQ	reported	0.835	0.007
PDRF_NOQ	shuffled	0.835	0.007
PDRF_NOQ	silent	0.835	0.007

Table S37: Complete grouping distribution. PDRF wins count paired seed differences within a grouping.

Grouping	Method	Macro-AUROC	Seed SD	PDRF–CAGF	PDRF wins
consecutive	CAGF	0.803	0.028		
consecutive	PDRF	0.826	0.007	0.023	3/5
interleaved	CAGF	0.824	0.017		
interleaved	PDRF	0.829	0.018	0.006	2/5
independent_16	CAGF	0.781	0.017		
independent_16	PDRF	0.813	0.007	0.032	5/5
random_01	CAGF	0.833	0.007		
random_01	PDRF	0.837	0.015	0.005	2/5
random_02	CAGF	0.815	0.019		
random_02	PDRF	0.832	0.012	0.017	4/5
random_03	CAGF	0.840	0.026		
random_03	PDRF	0.856	0.009	0.016	4/5
random_04	CAGF	0.837	0.012		
random_04	PDRF	0.849	0.009	0.012	3/5
random_05	CAGF	0.820	0.030		
random_05	PDRF	0.842	0.025	0.022	3/5
random_06	CAGF	0.840	0.018		
random_06	PDRF	0.856	0.010	0.016	3/5
random_07	CAGF	0.820	0.015		
random_07	PDRF	0.842	0.011	0.022	4/5
random_08	CAGF	0.812	0.031		
random_08	PDRF	0.842	0.009	0.030	3/5
random_09	CAGF	0.847	0.033		
random_09	PDRF	0.860	0.016	0.013	3/5
random_10	CAGF	0.812	0.013		
random_10	PDRF	0.850	0.013	0.038	5/5

Table S38: Leave-one-group-out degradation training. Values are means across five seeds.

Excluded/test group	Method	Macro-AUROC	Accuracy
1	CAGF	0.749	0.447
1	PDRF	0.782	0.505
2	CAGF	0.771	0.482
2	PDRF	0.790	0.519
3	CAGF	0.752	0.461
3	PDRF	0.797	0.515
4	CAGF	0.743	0.454
4	PDRF	0.789	0.502

Table S39: Bound, weighting-strength and ranking-weight sensitivity. Saturation denotes the affected fraction with $|r| \geq 0.95B$.

Setting	Macro-AUROC	Accuracy	Saturation
beta_0.1	0.821	0.531	0.470
beta_0.25	0.835	0.543	0.622
beta_0.5	0.846	0.555	0.690
beta_0.75	0.849	0.558	0.716
bound_1	0.821	0.535	0.780
bound_2	0.831	0.541	0.660
bound_3	0.835	0.543	0.622
bound_4	0.836	0.541	0.530
bound_5	0.835	0.545	0.397
rank_weight_0.05	0.825	0.539	0.247
rank_weight_0.15	0.835	0.543	0.622
rank_weight_0.3	0.835	0.542	0.599

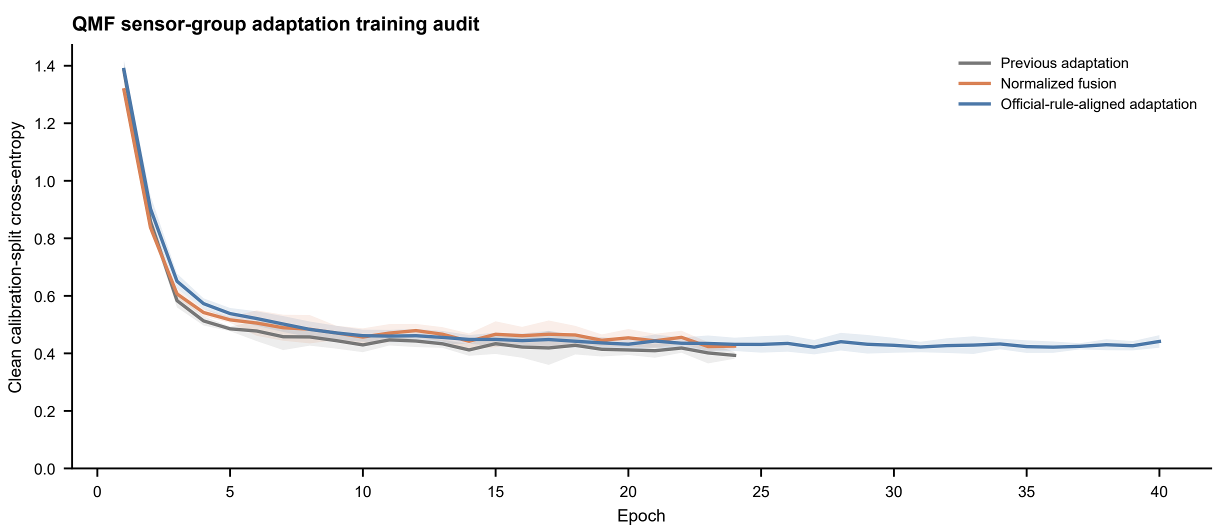

Figure S1: **QMF sensor-group training audit.** Mean clean calibration-split cross-entropy by epoch for the previous adaptation, normalized fusion and official-rule-aligned adaptation. Ribbons are standard deviations across available repeated fits; epochs with fewer than three active fits after early stopping are omitted. The calibration split, not the held-out test set, controls checkpoint selection.

Table S40: Hydraulic cooler and valve condition values and split counts.

Split	Condition value	n
train	3	288
train	20	288
train	100	293
selection	3	72
selection	20	72
selection	100	73
calibration	3	48
calibration	20	48
calibration	100	49
test	3	72
test	20	72
test	100	74

Split	Condition value	n
train	73	216
train	80	216
train	90	216
train	100	221
selection	73	54
selection	80	54
selection	90	54
selection	100	55
calibration	73	36
calibration	80	36
calibration	90	36
calibration	100	37
test	73	54
test	80	54
test	90	54
test	100	56

Table S41: Hydraulic pump and accumulator condition values and split counts.

Split	Condition value	n
train	0	293
train	1	288
train	2	288
selection	0	73
selection	1	72
selection	2	72
calibration	0	49
calibration	1	48
calibration	2	48
test	0	74
test	1	72
test	2	72

Split	Condition value	n
train	90	221
train	100	216
train	115	216
train	130	216
selection	90	55
selection	100	54
selection	115	54
selection	130	54
calibration	90	37
calibration	100	36
calibration	115	36
calibration	130	36
test	90	56
test	100	54
test	115	54
test	130	54

Table S42: Paired hydraulic PDRF-NOQ minus CAGF macro-AUROC effects. The minimum attainable non-zero two-sided exact Wilcoxon value with five same-direction pairs is 0.0625.

Task	Fault	Mean AUROC difference	PDRF wins	Exact P
accumulator	silent_pressure	0.094	5/5	0.0625
accumulator	silent_vibration	0.143	5/5	0.0625
cooler	silent_pressure	0.041	5/5	0.0625
cooler	silent_vibration	0.040	5/5	0.0625
pump	silent_pressure	0.053	5/5	0.0625
pump	silent_vibration	0.068	5/5	0.0625
valve	silent_pressure	0.067	5/5	0.0625
valve	silent_vibration	0.178	5/5	0.0625

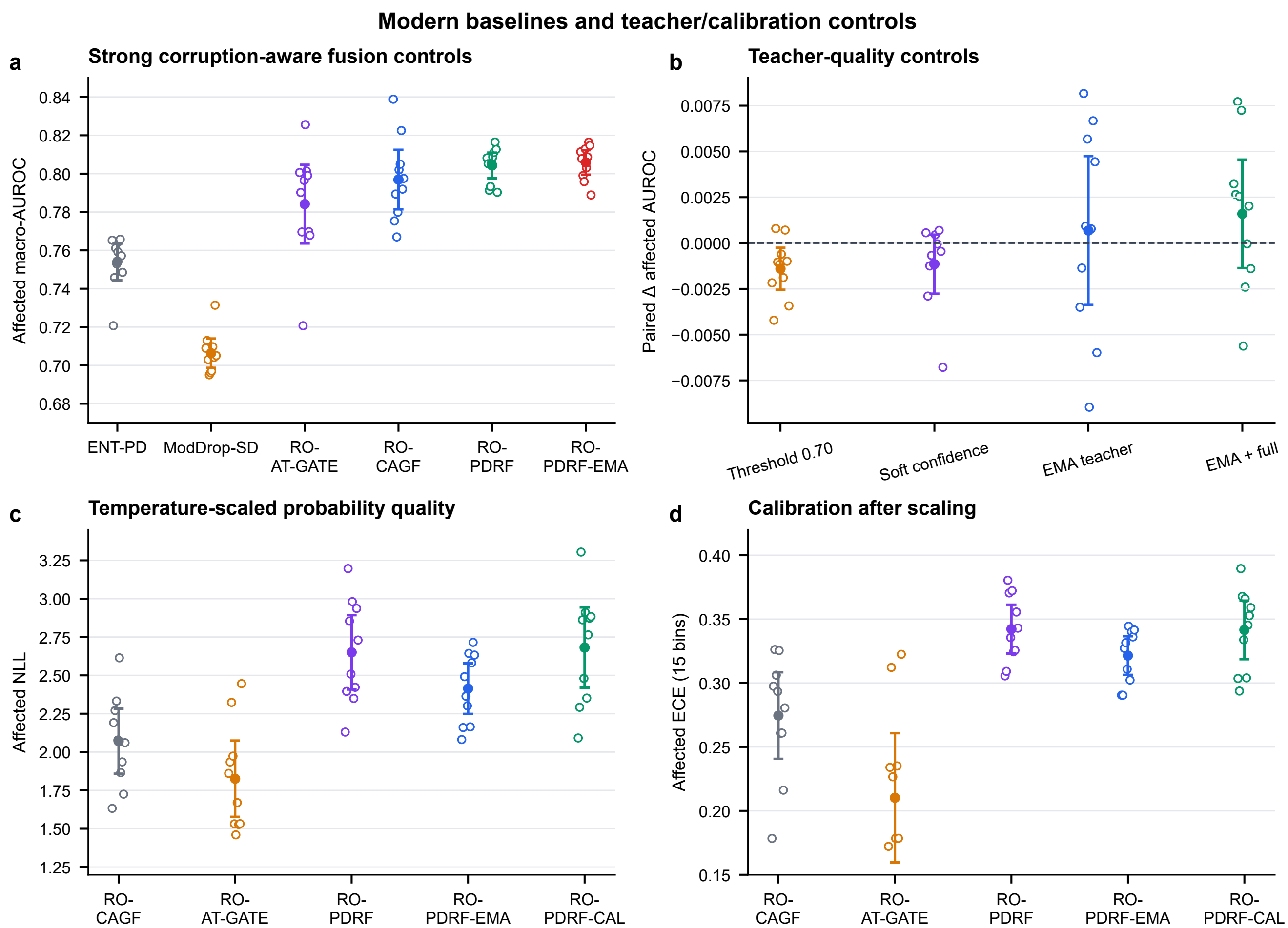

Figure S2: **Modern fusion and teacher/calibration controls.** **a**, Affected macro-AUROC for entropy weighting, modality-dropout self-distillation, parameter-matched attention gating, unrestricted recovery gating, bounded recovery fusion and its EMA extension. **b**, Paired affected-AUROC differences for hard-threshold, continuous-confidence and EMA teachers. **c,d**, Temperature-scaled affected NLL and ECE. Open circles are individual optimization seeds; filled circles are means and whiskers are 95% t intervals across the 10 repeated fits. The intervals describe optimization variation on one fixed test fault, not device replication.

Table S50: Original matched computational-cost audit on a 13th Gen Intel Core i5-13500H using PyTorch in CPU mode. Means and standard deviations are across three fixed seeds. Training time includes checkpoint selection; inference excludes data loading.

Method	Parameters	Training time (s)	Epochs	Inference (ms/observation)
CAGF	26,588	10.2 ± 1.1	21.0	0.0013 ± 0.0001
ENT-PD	17,560	9.0 ± 2.4	19.0	0.0013 ± 0.0003
PDRF	19,740	13.2 ± 3.9	22.3	0.0012 ± 0.0002
RO-PDRF-Full	19,740	16.6 ± 4.7	22.3	0.0016 ± 0.0001

Table S51: Expanded measured cost audit for the principal QMF, attention, Lite, Full, EMA and teacher-safeguard runs. Times are descriptive wall-clock measurements from the same CPU software environment; early stopping creates different epoch ranges. Inference excludes data loading.

Method	Parameters	Training time, mean ± SD (s)	Epochs represented	Inference (ms/observation)
QMF-PD	17,560	16.0 ± 3.5	24–47	0.0010
RO-AT-GATE	26,809	10.9 ± 3.6	13–30	0.0018
RO-CAGF	26,588	14.1 ± 3.0	15–26	0.0020
RO-PDRF-Lite	19,740	9.2 ± 2.7	18–41	0.0008
RO-PDRF-Full	19,740	22.7 ± 5.6	18–37	0.0020
RO-PDRF-EMA	19,740	23.8 ± 4.7	23–41	0.0017
RO-PDRF-Full-CRA	19,740	17.6 ± 5.2	18–41	0.0014
RO-PDRF-Full-ECA	19,740	19.3 ± 5.2	19–41	0.0014
CE-conflict training audit	19,740	22.0 ± 6.1	18–41	0.0018

Lite and Full have the same 19,740 trainable parameters because their difference is objective-only. Lite required 9.2 s mean training time versus 22.7 s for Full, while one-pass inference remained below 0.002 ms per observation. QMF-PD, RO-AT-GATE and RO-PDRF-EMA required 16.0, 10.9 and 23.8 s, respectively. These measurements support Lite as the simpler practical default but are not hardware-independent complexity bounds.

13 Fault-type, realization and EMA-decay analysis

The expanded chemical audit crosses Gaussian noise, a featurewise signed offset, acquisition-ordered monotone drift and stuck-at replacement with eight fixed intervention realizations and five fitted seeds. Scale, affected prevalence and random group missingness remain 3, 0.40 and 0.20. Within a realization, affected rows and missingness masks are shared across fault types; all intervention values are fixed across methods. The bootstrap resamples fault type, realization within type and the crossed optimization-seed cluster, preserving that one fitted seed is evaluated under every controlled intervention. These factors describe one chemical instrument, not independent hardware.

Table S52: Affected-subset discrimination and probability quality across four controlled fault mechanisms, eight realizations and five fitted seeds. Intervals for the EMA99 AUROC contrast use 5,000 crossed-factor bootstrap replicates.

Fault type	RO-CAGF AUROC	RO-PDRF-Full AUROC	EMA99 AUROC	EMA99 NLL	EMA99 ECE	Δ AUROC vs Full
Drift	0.791	0.826	0.829	2.139	0.285	+0.003 [-0.003, +0.010]
Gaussian	0.792	0.803	0.808	2.231	0.299	+0.005 [+0.002, +0.008]
Offset	0.786	0.816	0.819	2.382	0.319	+0.003 [-0.003, +0.011]
Stuck-at	0.778	0.811	0.816	2.247	0.298	+0.004 [-0.002, +0.012]

Table S53: Per-fault architecture effects. Intervals resample fault realization and the crossed optimization-seed cluster within each controlled mechanism. The four-type aggregate resamples only four top-level clusters and is therefore descriptive.

Controlled fault type	RO-CAGF AUROC	RO-PDRF-Full AUROC	Difference (95% hierarchical interval)
Gaussian	0.792	0.803	+0.011 [-0.009, +0.028]
Offset	0.786	0.816	+0.030 [+0.003, +0.057]
Drift	0.791	0.826	+0.035 [+0.012, +0.057]
Stuck-at	0.778	0.811	+0.034 [+0.003, +0.062]
Four-type descriptive aggregate	–	–	+0.027 [+0.002, +0.051]

Table S54: Leave-one-fault-type-out sensitivity for the RO-PDRF-Full–RO-CAGF affected-AUROC difference. Each row has only three remaining top-level fault clusters; intervals are descriptive robustness summaries, not device-generalization intervals.

Omitted controlled fault type	Remaining top-level clusters	RO-PDRF-Full–RO-CAGF AUROC (95% interval)
Gaussian	3	+0.033 [+0.007, +0.056]
Offset	3	+0.027 [+0.001, +0.052]
Drift	3	+0.025 [-0.002, +0.052]
Stuck-at	3	+0.025 [+0.000, +0.050]

Across all four mechanisms, RO-PDRF-Full exceeded RO-CAGF by a descriptive mean AUROC difference of +0.0274 (95% crossed-factor interval +0.0019 to +0.0512). The per-fault effects were +0.0112, +0.0299, +0.0351 and +0.0336 for Gaussian, offset, drift and stuck-at interventions. Every leave-one-type-out mean remained positive, although the interval after omitting drift crossed zero. Only four fault mechanisms define the top level, so these results support controlled-fault breadth without estimating a population of natural failure types. Across the same design, RO-PDRF-EMA changed Full-model AUROC by +0.0038 (−0.0007 to +0.0091), NLL by −0.283 (−0.535 to −0.094) and ECE by −0.0228 (−0.0396 to −0.0085).

Table S55: EMA-decay sensitivity under the principal Gaussian realization. Values are means ± standard deviations across five fitted seeds.

EMA decay	Affected AUROC	NLL	ECE
0.95	0.807 ± 0.010	2.550 ± 0.214	0.326 ± 0.022
0.99	0.806 ± 0.008	2.264 ± 0.178	0.301 ± 0.020
0.995	0.805 ± 0.010	2.205 ± 0.077	0.302 ± 0.018

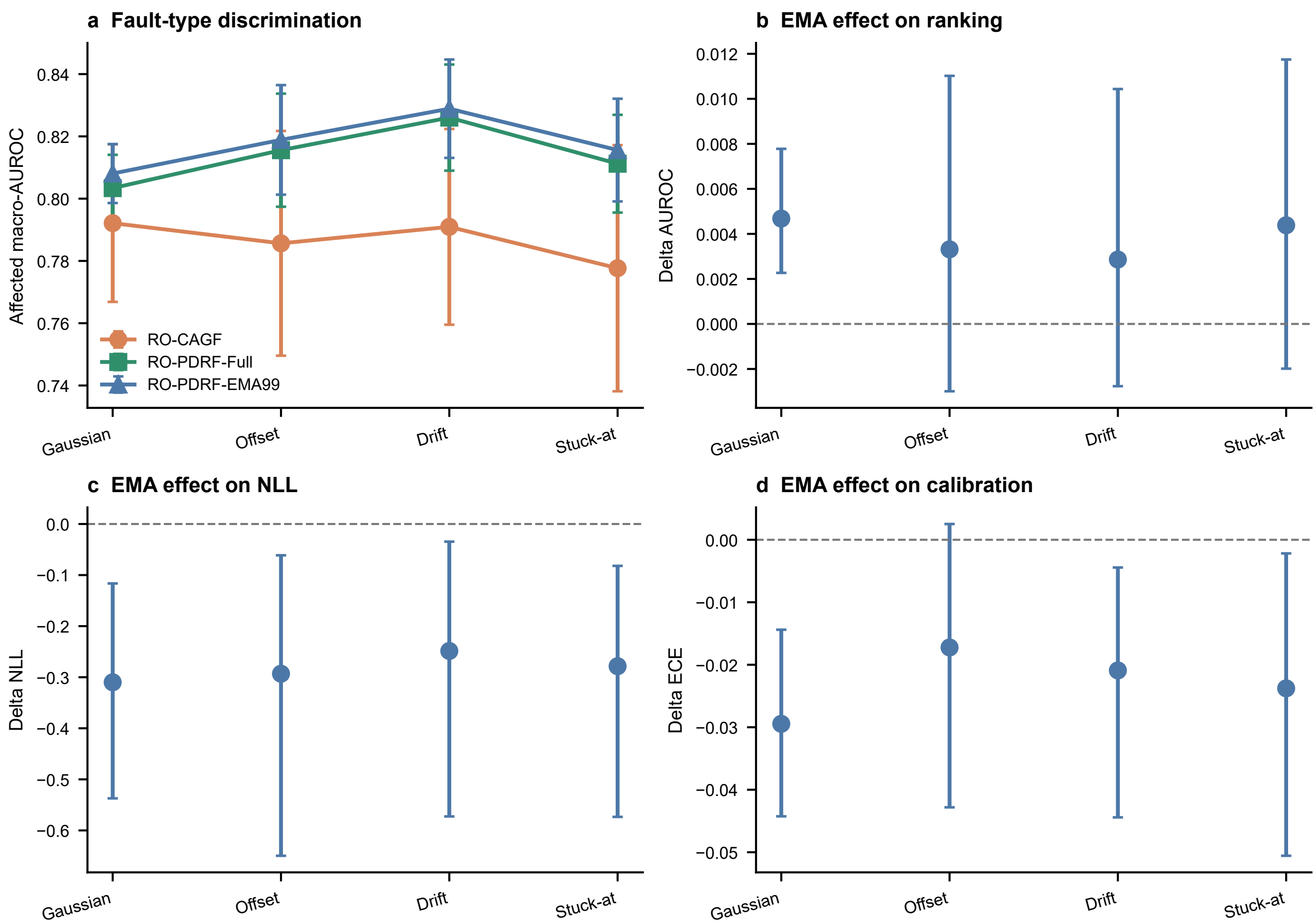

Figure S3: **Fault-type and EMA audit.** **a**, affected macro-AUROC by controlled mechanism. **b–d**, RO-PDRF-EMA minus RO-PDRF-Full effects for AUROC, NLL and ECE. Error bars use realization and crossed-seed resampling within each fault type. All interventions are repeated on one chemical instrument.

14 Score-geometry sensitivity

The following three-seed audit changes one parameter at a time around RO-PDRF-Full. It is exploratory and was not used to replace the formal setting. A larger bound lowers outer-5% occupancy and improves affected AUROC over this grid; larger β improves discrimination but does not monotonically reduce saturation. Ranking weight also changes saturation materially. No single hyperparameter is therefore justified by AUROC alone.

Table S56: Joint score-saturation, discrimination and calibration sensitivity. Values are means across three fixed development seeds.

Setting	Saturation	All AUROC	Affected AUROC	NLL	ECE
B=2	0.605	0.837	0.795	2.103	0.271
B=3	0.556	0.839	0.801	2.052	0.268
B=4	0.524	0.841	0.808	1.989	0.263
B=5	0.420	0.845	0.814	2.019	0.262
beta=0.1	0.436	0.831	0.783	2.150	0.274
beta=0.25	0.556	0.839	0.801	2.052	0.268
beta=0.5	0.516	0.850	0.824	1.932	0.249
beta=0.75	0.612	0.853	0.832	1.905	0.253
rank=0.05	0.294	0.832	0.790	2.150	0.272
rank=0.15	0.556	0.839	0.801	2.052	0.268
rank=0.3	0.650	0.840	0.804	2.113	0.274

15 Hydraulic no-interpolation audit

The principal hydraulic representation linearly resamples each physical-group summary vector to length 48. The no-interpolation audit instead retains every native summary value in its original order and zero-pads only the unused tail. Both representations use identical samples, splits, faults and seeds.

Table S57: Affected-subset macro-AUROC under controlled hydraulic pressure faults. Difference is RO-PDRF-NOQ minus CAGF.

Representation	Task	CAGF	RO-PDRF-Full	Difference
Linear resampling	Accumulator	0.608	0.693	+0.085
Linear resampling	Cooler	0.949	1.000	+0.051
Linear resampling	Pump	0.871	0.921	+0.050
Linear resampling	Valve	0.697	0.684	-0.013
No interpolation	Accumulator	0.575	0.710	+0.135
No interpolation	Cooler	0.919	1.000	+0.081
No interpolation	Pump	0.787	0.911	+0.124
No interpolation	Valve	0.681	0.656	-0.025

16 Real-operational AHU direct-diagnosis boundary analysis

The external field audit uses one year of hourly operational records from each of three buildings and only the repository's expert annotations for normal operation, return-air temperature sensor fault and supply-air temperature sensor fault. No test fault is synthesized. Table S58 reports the chronological partitions after exclusion of conflicting duplicate AHU–time keys and collapse of same-label repeats. The complete test periods are retained without class subsampling.

For leave-one-building-out analysis, models train on two buildings and hold out the third. Shift diagnostics include standardized mean, variance, quantile, missingness and outlier differences per feature; Kolmogorov–Smirnov and Wasserstein distances; linear and radial-basis maximum mean discrepancy; covariance discrepancy; a source-trained PCA projection; and class/subtype priors. A reversal audit reports AUROC from $1 - p(\text{fault})$ to distinguish absent ranking information from approximately inverted ranking. Source-only controls use source normalization and include EF-PD, RO-PDRF-Full-NOQ and GroupDRO-EF-PD with building as the source group. Target normalization recomputes location and scale from unlabeled target features, while CORAL aligns source covariance to unlabeled target covariance. These two regimes are unsupervised target adaptation, not source-only transfer.

Five-member probabilities are averaged for risk–coverage analysis. Split-conformal sets use calibration nonconformity $1 - p_y$ at nominal error levels 0.05, 0.10 and 0.20. Empirical coverage, mean set size, singleton rate and singleton accuracy are diagnostics only because chronological and cross-building shift violate simple exchangeability.

Table S58: Chronological AHU split counts used in the field validation.

Building	Split	Records	Normal	Sensor fault
Auditorium	Train	7,658	6,000	1,658
Auditorium	Selection	1,728	1,500	228
Auditorium	Calibration	1,524	1,500	24
Auditorium	Test	9,354	9,126	228
Hospital	Train	11,300	6,000	5,300
Hospital	Selection	2,675	1,500	1,175
Hospital	Calibration	2,318	1,500	818
Hospital	Test	9,362	7,867	1,495
Office	Train	11,727	6,000	5,727
Office	Selection	3,000	1,500	1,500
Office	Calibration	1,755	1,500	255
Office	Test	19,115	18,769	346

Table S59: Class and native sensor-fault-subtype distribution in each untouched chronological test period.

Building	Normal	Return-air fault	Supply-air fault	Total
Auditorium	9,126	179	49	9,354
Hospital	7,867	1,487	8	9,362
Office	18,769	293	53	19,115

The hospital test period is qualitatively different from the other two buildings: it contains 1,487 return-air faults but only eight supply-air faults. Metrics are therefore reported by building and subtype rather than pooled across hourly observations.

Table S63 treats the 15 building–seed combinations as matched repeated fits. These tests describe optimization consistency across the three building-specific tasks; they do not treat hourly records as independent retraining replicates. EF-PD and HGB-current exceed RO-PDRF-NOQ in all 15 pairs. HGB-lag does not uniformly improve HGB-current, showing that these fixed lag summaries are a task-specific baseline rather than a surrogate for every sequence architecture.

Empirical coverage is frequently below the nominal 0.90 target, most notably in the auditorium. This is evidence against claiming a distribution-free field guarantee: chronological changes violate the exchangeability condition needed for the elementary split-conformal construction. Confidence-based abstention can still reduce selective error, but confidence ordering and class-specific recall must be audited together.

Table S60: Detailed chronological AHU results. Values are means across five matched seeds. Fault precision, recall and F1 treat either native temperature-sensor-fault subtype as the positive class.

Building	Method	AUROC	AUPRC	Macro-F1	Bal. Acc.	Fault Prec.	Fault Recall	Fault F1
Auditorium	EF-PD	0.996	0.957	0.901	0.913	0.793	0.832	0.807
Auditorium	HGB-current	0.993	0.932	0.854	0.941	0.598	0.896	0.717
Auditorium	HGB-lag	0.992	0.925	0.863	0.871	0.720	0.749	0.733
Auditorium	ENT-PD	0.980	0.833	0.755	0.893	0.395	0.818	0.529
Auditorium	CAGF	0.990	0.907	0.832	0.861	0.666	0.733	0.674
Auditorium	PDRF-NOQ	0.985	0.812	0.695	0.954	0.271	0.973	0.424
Auditorium	RO-PDRF-Full-NOQ	0.984	0.809	0.698	0.943	0.276	0.947	0.428
Hospital	EF-PD	0.990	0.983	0.947	0.926	0.969	0.857	0.909
Hospital	HGB-current	0.978	0.968	0.946	0.921	0.983	0.845	0.909
Hospital	HGB-lag	0.964	0.958	0.941	0.911	0.993	0.822	0.899
Hospital	ENT-PD	0.958	0.950	0.935	0.909	0.965	0.824	0.889
Hospital	CAGF	0.912	0.928	0.943	0.913	0.995	0.828	0.903
Hospital	PDRF-NOQ	0.936	0.939	0.941	0.909	1.000	0.817	0.899
Hospital	RO-PDRF-Full-NOQ	0.931	0.937	0.940	0.908	1.000	0.815	0.898
Office	EF-PD	0.998	0.945	0.816	0.987	0.481	0.995	0.642
Office	HGB-current	0.996	0.897	0.756	0.984	0.360	1.000	0.529
Office	HGB-lag	0.996	0.882	0.756	0.984	0.359	1.000	0.528
Office	ENT-PD	0.975	0.727	0.786	0.869	0.479	0.753	0.581
Office	CAGF	0.984	0.845	0.770	0.967	0.422	0.971	0.559
Office	PDRF-NOQ	0.968	0.709	0.630	0.950	0.180	0.983	0.304
Office	RO-PDRF-Full-NOQ	0.966	0.704	0.620	0.914	0.169	0.912	0.285

A deliberately harder leave-one-building-out audit trains on two buildings and evaluates the third without target-building adaptation. The five-seed results in Table S65 show substantial domain shift, especially for the hospital. This audit is not used to select the temporal protocol or to claim device-independent transfer; it identifies a deployment boundary that requires explicit domain adaptation.

The failure is concentrated rather than hidden by an overall average. RO-PDRF-NOQ retained AUROC 0.984 ± 0.000 for the held-out auditorium and 0.887 ± 0.006 for the held-out office, but fell to 0.298 ± 0.005 for the hospital. ENT-PD and PDRF-NOQ failed similarly on the hospital, whereas EF-PD retained 0.983 ± 0.003. The learned bounded-score pathway therefore did not align across the hospital domain.

The expanded audit diagnoses this boundary instead of assuming that one adaptation is sufficient. Reversed hospital RO-PDRF-Full-NOQ probabilities yield AUROC 0.702 ± 0.005, revealing approximately inverted ranking. Target normalization and CORAL improve the original AUROC only to 0.356 ± 0.005 and 0.367 ± 0.006 and therefore do not repair it. GroupDRO-EF-PD remains strong at 0.982 ± 0.004 but does not improve on source-normalized EF-PD. Full feature-level shifts, PCA coordinates, prediction files and risk–coverage traces are supplied as source data.

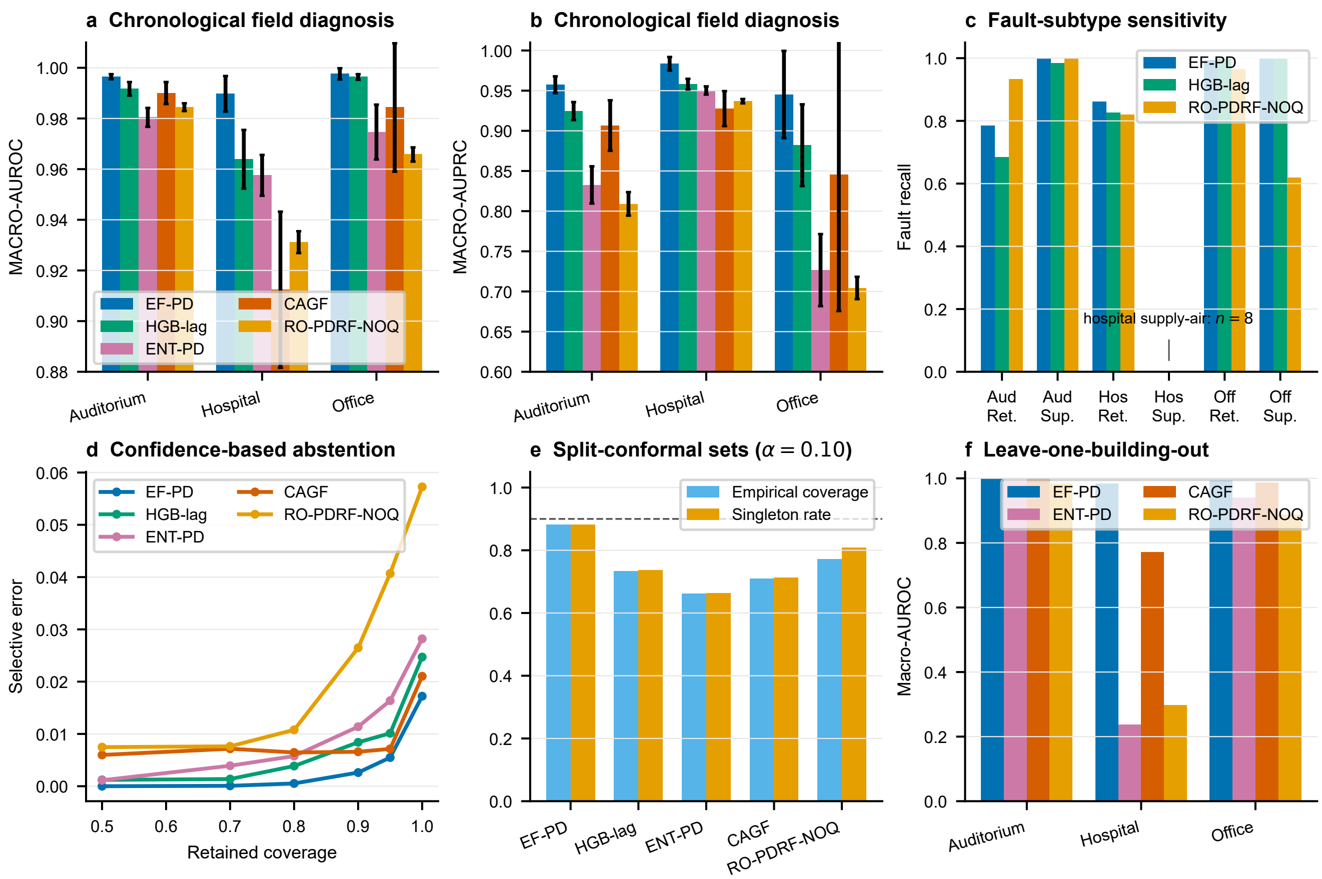

Figure S4: **Operational AHU diagnosis and deployment boundaries.** **a,b**, Building-specific chronological AUROC and AUPRC. **c**, Return- and supply-air fault recall, with the rare hospital supply-air subset identified. **d**, Selective error as lower-confidence predictions are rejected. **e**, Empirical split-conformal coverage and singleton rate at nominal 90% coverage. **f**, leave-one-building-out AUROC without target-building adaptation. Error bars in **a**–**c** show standard deviations across five seeds.

Table S61: Recall for the two native expert-annotated fault subtypes. The class counts in Table S59 govern interpretation, especially for the eight hospital supply-air faults.

Building	Method	Fault subtype	Recall	Number of seeds
Auditorium	EF-PD	Return-air	0.785 ± 0.097	5
Auditorium	EF-PD	Supply-air	1.000 ± 0.000	5
Auditorium	HGB-current	Return-air	0.873 ± 0.024	5
Auditorium	HGB-current	Supply-air	0.984 ± 0.009	5
Auditorium	HGB-lag	Return-air	0.685 ± 0.045	5
Auditorium	HGB-lag	Supply-air	0.984 ± 0.027	5
Auditorium	ENT-PD	Return-air	0.783 ± 0.108	5
Auditorium	ENT-PD	Supply-air	0.947 ± 0.031	5
Auditorium	CAGF	Return-air	0.667 ± 0.171	5
Auditorium	CAGF	Supply-air	0.976 ± 0.037	5
Auditorium	PDRF-NOQ	Return-air	0.965 ± 0.009	5
Auditorium	PDRF-NOQ	Supply-air	1.000 ± 0.000	5
Auditorium	RO-PDRF-Full-NOQ	Return-air	0.933 ± 0.012	5
Auditorium	RO-PDRF-Full-NOQ	Supply-air	1.000 ± 0.000	5
Hospital	EF-PD	Return-air	0.862 ± 0.020	5
Hospital	EF-PD	Supply-air	0.000 ± 0.000	5
Hospital	HGB-current	Return-air	0.846 ± 0.003	5
Hospital	HGB-current	Supply-air	0.675 ± 0.381	5
Hospital	HGB-lag	Return-air	0.827 ± 0.001	5
Hospital	HGB-lag	Supply-air	0.000 ± 0.000	5
Hospital	ENT-PD	Return-air	0.828 ± 0.002	5
Hospital	ENT-PD	Supply-air	0.000 ± 0.000	5
Hospital	CAGF	Return-air	0.832 ± 0.024	5
Hospital	CAGF	Supply-air	0.000 ± 0.000	5
Hospital	PDRF-NOQ	Return-air	0.822 ± 0.002	5
Hospital	PDRF-NOQ	Supply-air	0.000 ± 0.000	5
Hospital	RO-PDRF-Full-NOQ	Return-air	0.820 ± 0.005	5
Hospital	RO-PDRF-Full-NOQ	Supply-air	0.000 ± 0.000	5
Office	EF-PD	Return-air	0.994 ± 0.008	5
Office	EF-PD	Supply-air	1.000 ± 0.000	5
Office	HGB-current	Return-air	1.000 ± 0.000	5
Office	HGB-current	Supply-air	1.000 ± 0.000	5
Office	HGB-lag	Return-air	1.000 ± 0.000	5
Office	HGB-lag	Supply-air	1.000 ± 0.000	5
Office	ENT-PD	Return-air	0.792 ± 0.071	5
Office	ENT-PD	Supply-air	0.532 ± 0.117	5
Office	CAGF	Return-air	0.966 ± 0.050	5
Office	CAGF	Supply-air	1.000 ± 0.000	5
Office	PDRF-NOQ	Return-air	0.992 ± 0.002	5
Office	PDRF-NOQ	Supply-air	0.932 ± 0.077	5
Office	RO-PDRF-Full-NOQ	Return-air	0.965 ± 0.020	5
Office	RO-PDRF-Full-NOQ	Supply-air	0.619 ± 0.130	5

Table S62: Confusion counts for fixed five-member probability ensembles on the untouched chronological test periods. Either native sensor-fault subtype is positive.

Building	Method	TN	FP	FN	TP
Auditorium	EF-PD	9,080	46	41	187
Auditorium	HGB-current	9,000	126	23	205
Auditorium	HGB-lag	9,061	65	55	173
Auditorium	ENT-PD	8,824	302	36	192
Auditorium	CAGF	9,072	54	70	158
Auditorium	RO-PDRF-Full-NOQ	8,557	569	11	217
Hospital	EF-PD	7,853	14	214	1,281
Hospital	HGB-current	7,848	19	232	1,263
Hospital	HGB-lag	7,860	7	267	1,228
Hospital	ENT-PD	7,866	1	262	1,233
Hospital	CAGF	7,866	1	273	1,222
Hospital	RO-PDRF-Full-NOQ	7,867	0	274	1,221
Office	EF-PD	18,426	343	1	345
Office	HGB-current	18,166	603	0	346
Office	HGB-lag	18,157	612	0	346
Office	ENT-PD	18,458	311	78	268
Office	CAGF	18,378	391	2	344
Office	RO-PDRF-Full-NOQ	17,259	1,510	29	317

Table S63: Paired AHU macro-AUROC contrasts for RO-PDRF-NOQ across five seeds and three buildings.

Contrast	Mean Δ AUROC	Wins	Losses	Exact P
RO-PDRF-Full-NOQ vs. PDRF-NOQ	-0.0023	2	13	0.0006
RO-PDRF-Full-NOQ vs. CAGF	-0.0018	6	9	0.8040
RO-PDRF-Full-NOQ vs. ENT-PD	-0.0104	5	10	0.0479
RO-PDRF-Full-NOQ vs. EF-PD	-0.0341	0	15	0.0001
RO-PDRF-Full-NOQ vs. UF-PD	-0.0066	7	8	0.1514
RO-PDRF-Full-NOQ vs. HGB-current	-0.0285	0	15	0.0001
RO-PDRF-Full-NOQ vs. HGB-lag	-0.0235	0	15	0.0001

Table S64: Split-conformal diagnostics at nominal error level $\alpha = 0.10$. Calibration uses the chronological calibration interval and evaluation uses the later test interval.

Building	Method	Coverage	Mean set size	Singleton rate	Singleton accuracy
Auditorium	CAGF	0.674	0.674	0.674	1.000
Auditorium	EF-PD	0.832	0.832	0.832	1.000
Auditorium	ENT-PD	0.548	0.548	0.548	1.000
Auditorium	HGB-lag	0.627	0.627	0.627	1.000
Auditorium	RO-PDRF-Full-NOQ	0.525	0.525	0.525	1.000
Hospital	CAGF	0.586	0.597	0.597	0.981
Hospital	EF-PD	0.983	1.017	0.983	0.983
Hospital	ENT-PD	0.520	0.523	0.523	0.995
Hospital	HGB-lag	0.637	0.639	0.639	0.996
Hospital	RO-PDRF-Full-NOQ	0.875	0.902	0.902	0.970
Office	CAGF	0.868	0.868	0.868	1.000
Office	EF-PD	0.829	0.830	0.830	0.999
Office	ENT-PD	0.916	0.921	0.921	0.994
Office	HGB-lag	0.939	0.944	0.944	0.995
Office	RO-PDRF-Full-NOQ	0.918	0.998	0.998	0.920

Table S65: Leave-one-building-out macro-AUROC averaged across five fixed seeds.

Held-out building	EF-PD	ENT-PD	CAGF	PDRF-NOQ	RO-PDRF-Full-NOQ
Auditorium	1.000 ± 0.000	0.990 ± 0.001	1.000 ± 0.000	0.983 ± 0.001	0.984 ± 0.000
Hospital	0.983 ± 0.003	0.238 ± 0.012	0.772 ± 0.167	0.302 ± 0.004	0.298 ± 0.005
Office	0.994 ± 0.002	0.941 ± 0.007	0.986 ± 0.002	0.884 ± 0.006	0.887 ± 0.006

Table S66: Multivariate leave-one-building-out shift distances. Values are descriptive distances computed without target labels.

Held-out building	Linear MMD	RBF MMD	CORAL distance
Auditorium	1.448	0.133	0.00062
Hospital	3.102	0.198	0.00104
Office	1.354	0.073	0.00106

Table S67: Largest hospital feature shifts under source normalization, including location, dispersion, missingness and outlier diagnostics.

Feature	Standardized mean difference	KS	Wasserstein	Target outside source 1–99%
cooling pump	+2.52	0.98	0.71	0.0%
set point temperature	+1.24	0.61	2.91	0.0%
return temperature	+1.01	0.45	2.93	0.4%
supply fan	+0.63	0.34	0.29	0.0%
supply air temperature	+0.60	0.35	2.95	1.4%
heating supply temperature	+0.58	0.38	6.81	15.1%
valve position	-0.47	0.17	10.30	0.0%
cooling supply temperature	-0.12	0.44	4.03	19.1%

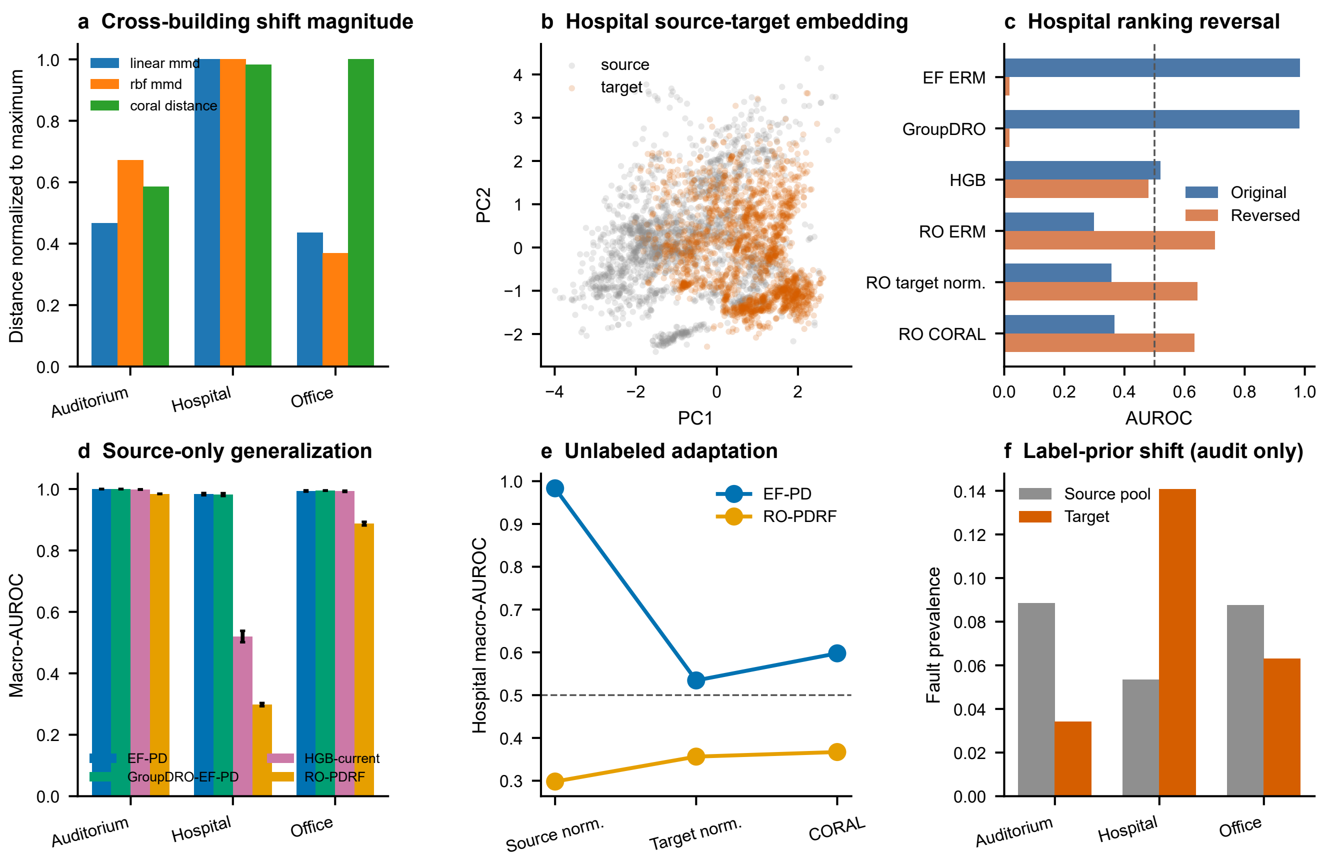

Figure S5: **Full cross-building shift and adaptation audit.** Panels report source-trained PCA geometry, multivariate distances, feature-level shifts, source/target class and subtype priors, five-seed direct-diagnosis AUROC, hospital reversed-score AUROC and selective-risk behaviour. Target normalization and CORAL use unlabeled target features and are not zero-shot methods.

17 Protocol reconciliation and recovery-objective controls

The legacy unified assignment protocol uses the clean-view teacher, fixed chemical test-fault seed 70001 and the same 10 optimization seeds for CAGF, RO-CAGF, PDRF and RO-PDRF-Full. An original recovery/calibration run used the same test fault but a label-assisted teacher for the bounded recovery model. A superseded factorial rerun used a different fixed test-fault instance (seed 85000). The latter difference changed the assigned rows and explains why nominally identical legacy source-data identifiers previously had different values. Table S68 records every relevant protocol field. The revised main paper reuses the unified fitted models but restricts evaluation to fault-applied-and-available rows.

Table S68: Reconciliation of the original formal run, superseded factorial rerun and legacy unified assignment protocol. CE denotes cross-entropy.

Protocol field	Original formal run	Superseded rerun	Unified headline
Purpose	recovery/calibration audit	initial 2×2 audit	sole main-paper protocol
Recovery teacher	label-assisted best view	label-assisted best view	clean view
Chemical seeds	101–110	101–110	101–110
Data split indices	batches 1–6 / 7 / 8–10	same	same
Initialization	PyTorch seed 101–110	same	same
Test-fault seed	70001	85000	70001
Fault / group	Gaussian scale 3 / group 1	Gaussian scale 3 / group 1	Gaussian scale 3 / group 1
Affected / missing	40% / 20%	40% / 20%	40% / 20%
Quality at test	neutral ($q = 1$)	neutral ($q = 1$)	neutral ($q = 1$)
Checkpoint	natural selection CE	natural selection CE	natural selection CE
Training / inference paths	3 / 1	3 / 1	3 / 1
Evaluation subset	fixed affected rows	different fixed affected rows	fixed affected rows
Status	supplementary sensitivity	archived / superseded	headline

The recovery-objective-matched 2×2 comparison applies clean-view recovery distillation, Brier supervision and degraded-view ranking to both bounded PDRF and unrestricted CAGF. It remains architecture-specific because RO-PDRF-Full uses a response-interior penalty while RO-CAGF uses a gate KL penalty. The seed-level contrasts and CPU cost below complement the main paper. Exact tests remain descriptive, and the 17 chemical stress conditions are correlated sensitivity conditions rather than independent replications.

Table S69: Recovery-objective-matched chemical and hydraulic contrasts. Positive values favour the first named method.

Dataset	Contrast	Mean Δ AUROC (95% CI)	Wins	Losses	P
Chemical	RO-PDRF-Full - PDRF	+0.014 [+0.011, +0.016]	10	0	0.0020
Chemical	RO-CAGF - CAGF	+0.005 [-0.008, +0.017]	6	4	0.3750
Chemical	RO-PDRF-Full - RO-CAGF	+0.007 [-0.004, +0.018]	7	3	0.3223
Chemical	PDRF - CAGF	-0.001 [-0.014, +0.011]	5	5	0.9219
Chemical	interaction: bounded recovery gain - gate recovery gain	+0.008 [-0.003, +0.020]	7	3	0.2324
Hydraulic	RO-PDRF-Full - RO-CAGF	+0.044 [+0.016, +0.068]	7	1	0.0391
Hydraulic	RO-PDRF-Full - PDRF	+0.008 [-0.003, +0.021]	3	3	0.5625
Hydraulic	RO-CAGF - CAGF	+0.027 [+0.010, +0.047]	8	0	0.0078

The clean-view teacher is the formal model. Label-assisted best-view, affected-group-removal and confidence-selected teachers are sensitivity variants evaluated on their own prespecified fixed fault (seed 87001), separate from the seed-70001 headline fault. Their near-identical affected-subset performance within that matched sensitivity experiment shows that its result does not depend materially on teacher selection; it is not pooled with the headline comparison. A separate leave-one-loss-out audit removes recovery distillation, Brier, interior or degraded-AUC supervision one at a time from the clean-teacher objective under the headline fault.

The architecture-specific penalties are tested at weights 0, 0.001, 0.005 and 0.020. The complete 4×4 contrast matrix prevents the architecture conclusion from depending on one selected KL/interior pair.

Table S70: Recovery-objective-matched parameter counts and measured CPU costs. All four methods use three forward paths during training and one during inference.

Dataset	Method	Parameters	Train time (s)	Inference (ms/obs.)	Training paths
Chemical	CAGF	26,588	13.5 ± 4.0	0.0019 ± 0.0003	3
Chemical	PDRF	19,740	18.1 ± 5.9	0.0020 ± 0.0002	3
Chemical	RO-CAGF	26,588	14.1 ± 3.0	0.0020 ± 0.0002	3
Chemical	RO-PDRF-Full	19,740	22.7 ± 5.6	0.0020 ± 0.0002	3
Hydraulic	CAGF	37,844	7.5 ± 2.4	0.0116 ± 0.0011	3
Hydraulic	PDRF	29,300	10.4 ± 1.8	0.0115 ± 0.0010	3
Hydraulic	RO-CAGF	37,844	8.5 ± 3.0	0.0145 ± 0.0054	3
Hydraulic	RO-PDRF-Full	29,300	11.8 ± 1.9	0.0119 ± 0.0019	3

Table S71: Recovery-teacher audit across 10 matched chemical-array seeds. Clean-only is the formal teacher; best-CE uses training labels and is retained only as sensitivity.

Teacher selection	Aff. AUROC	Aff. AUPRC	Aff. NLL
clean only	0.823 ± 0.010	0.551 ± 0.018	2.425 ± 0.300
best ce	0.823 ± 0.010	0.552 ± 0.018	2.459 ± 0.305
removal only	0.822 ± 0.010	0.551 ± 0.018	2.457 ± 0.299
confidence selected	0.823 ± 0.010	0.550 ± 0.020	2.417 ± 0.306

Table S72: Leave-one-loss-out affected-AUROC contrasts. Positive full-minus-ablated differences favour the complete recovery objective.

Omitted term	Full-minus-ablated AUROC (95% CI)	Wins	Losses	P
Recovery	+0.010 [+0.008, +0.013]	10	0	0.0020
Brier	+0.002 [+0.001, +0.003]	9	1	0.0098
Interior	+0.001 [-0.000, +0.002]	6	4	0.2754
Auc	+0.002 [+0.001, +0.004]	8	2	0.0645

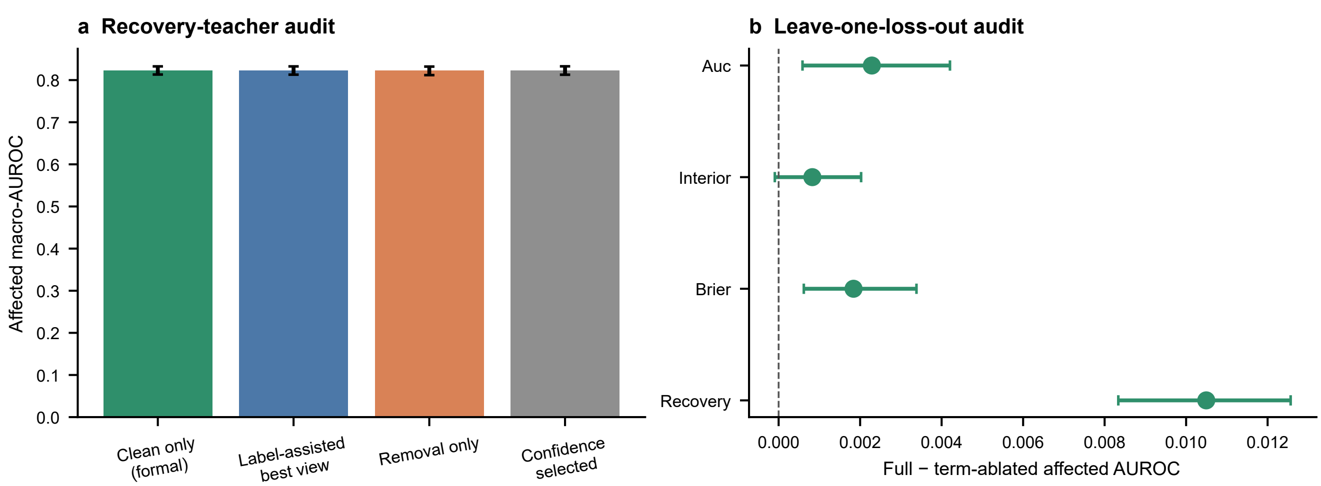

Figure S6: **Recovery-objective robustness.** **a**, affected macro-AUROC under four teacher rules. **b**, paired full-objective minus leave-one-loss-out effects. Error bars show standard deviations or paired 95% bootstrap intervals as indicated.

Table S73: Architecture-specific regularizer sensitivity under the unique chemical headline protocol.

Recovery model / regularizer weight	Aff. AUROC	SD
RO-CAGF KL=0	0.796	0.031
RO-CAGF KL=.001	0.795	0.028
RO-CAGF KL=.005	0.797	0.022
RO-CAGF KL=.020	0.799	0.018
RO-PDRF-Full interior=0	0.803	0.009
RO-PDRF-Full interior=.001	0.804	0.009
RO-PDRF-Full interior=.005	0.804	0.009
RO-PDRF-Full interior=.020	0.805	0.009

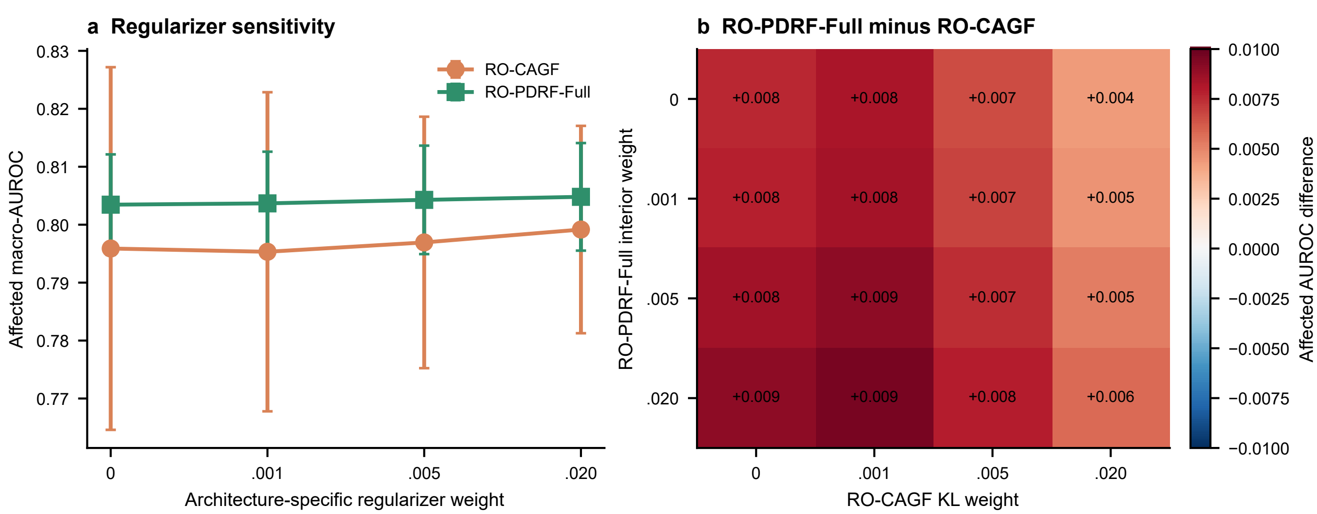

Figure S7: **Architecture-specific regularizer sensitivity.** **a**, affected macro-AUROC across gate-KL and bounded-response-interior weights. **b**, all 16 RO-PDRF-Full minus RO-CAGF regularizer combinations. Error bars show standard deviations across 10 matched seeds.

18 Consistency controls, clustered validation and teacher errors

The consistency ladder applies clean-view distillation alone to early fusion, uniform fusion, CAGF and PDRF before adding the complete recovery auxiliaries. The modality-dropout self-distillation control removes one available group per pair rather than injecting a Gaussian sensor corruption. These are principle-matched controls based on established modality-dropout and self-distillation ideas; they are not claimed as exact reproductions of a published end-to-end system. The confidence-thresholded PDRF control retains only teachers with maximum class probability at least 0.70.

Table S74: Clean-distillation ladder. Differences and 95% intervals are paired against clean distillation within architecture; exact two-sided Wilcoxon values compare matched optimization seeds.

Architecture	Model/training condition	Affected AUROC	Δ vs clean distillation (95% CI)	*P*
EF-PD	EF-PD	0.747 ± 0.006	-0.012 [-0.016, -0.007]	0.0020
EF-PD	EF-PD + clean distillation	0.758 ± 0.006	Reference	–
UF-PD	UF-PD	0.763 ± 0.010	-0.027 [-0.036, -0.018]	0.0039
UF-PD	UF-PD + clean distillation	0.791 ± 0.008	Reference	–
CAGF	CAGF	0.792 ± 0.027	-0.009 [-0.023, +0.006]	0.2754
CAGF	CAGF + clean distillation	0.800 ± 0.024	Reference	–
CAGF	RO-CAGF	0.797 ± 0.022	-0.003 [-0.012, +0.006]	0.6250
PDRF	PDRF	0.791 ± 0.010	-0.012 [-0.014, -0.009]	0.0020
PDRF	RO-PDRF-Lite	0.802 ± 0.008	Reference	–
PDRF	RO-PDRF-Full	0.804 ± 0.009	+0.002 [-0.001, +0.005]	0.1934

Eight chemical fault realizations use independently generated affected indices, missingness masks and Gaussian noise while remaining fixed across methods. Five fitted models are evaluated per realization. The hierarchical bootstrap samples fault realizations first and optimization seeds within realization second. Acquisition-batch summaries are retained separately; neither level represents independent hardware.

Table S75: Chemical affected-AUROC contrasts across eight fixed fault realizations. Intervals use the fault-realization–seed hierarchy.

Contrast	Mean Δ AUROC	95% hierarchical CI	Positive	Realizations
RO-PDRF-Full - RO-CAGF	+0.012	[+0.005, +0.019]	8	8
RO-PDRF-Full - PDRF	+0.012	[+0.010, +0.013]	8	8
RO-CAGF - CAGF	+0.011	[+0.006, +0.016]	8	8

Table S76: RO-PDRF-Full minus RO-CAGF affected-AUROC by acquisition batch across fault realizations and seeds.

Acquisition batch	Mean Δ	SD	Minimum	Maximum
10	+0.010	+0.022	-0.035	+0.036
8	+0.018	+0.033	-0.085	+0.094
9	+0.015	+0.026	-0.034	+0.073

The clean-teacher audit keeps each realization's availability mask identical between clean and faulted views. Correctness quadrants therefore distinguish preserved correctness, degradation-induced failure, recovery despite a wrong teacher and shared failure. A paired comparison with ordinary PDRF estimates how often a correction present in the non-distilled model is absent from RO-PDRF-Full when the latter's clean teacher is wrong. Because the two models are separately trained, this is an error-transfer association rather than a unit-level causal effect of the KL term.

Table S77: Clean-teacher and faulted-prediction correctness quadrants across five models and eight realizations.

Clean teacher	Faulted prediction	Rows	Fraction
wrong	wrong	25807	0.373
wrong	correct	3203	0.046
correct	wrong	8712	0.126
correct	correct	31468	0.455

Table S78: Teacher-confidence audit. Δp_y is RO-PDRF-Full minus PDRF true-class probability on the faulted observation.

Teacher-confidence quintile	Rows	Mean confidence	RO-base Δp_y	RO accuracy	Base accuracy
(0.225, 0.61]	13838	0.493	-0.007	0.245	0.248
(0.61, 0.82]	13838	0.717	0.001	0.306	0.304
(0.82, 0.953]	13839	0.897	0.016	0.448	0.422
(0.953, 0.995]	13837	0.979	0.026	0.680	0.646
(0.995, 1.0]	13838	0.999	0.018	0.828	0.810

19 Guarded hydraulic split and similarity audit

Within each native component-condition class, cycles remain in acquisition order and are assigned to contiguous training, checkpoint-selection, calibration and test blocks in 60/15/10/15% proportions. Five cycles at each boundary are discarded as guard bands. A global chronological split is not meaningful for these component tasks because the acquisition order contains state blocks: cooler training would omit the healthy state, while accumulator partitions would have non-overlapping states. The class-preserving blocked split therefore tests local sequence leakage without replacing the target with an unseen-class problem.

Table S79: RO-PDRF-Full minus RO-CAGF affected-AUROC under the guarded within-condition blocked split.

Task	Fault	Blocked-split Δ AUROC
accumulator	pressure	+0.013
accumulator	vibration	+0.009
cooler	pressure	+0.050
cooler	vibration	+0.053
pump	pressure	-0.033
pump	vibration	-0.012
valve	pressure	+0.032
valve	vibration	+0.010

Table S80: Earlier random-stratified hydraulic sensitivity analysis. These results are not the main hydraulic evidence because adjacent cycles can be allocated across splits.

Task	Fault	RO-CAGF	RO-PDRF-Full	Difference
Cooler	Pressure	0.944 ± 0.040	1.000 ± 0.000	+0.056
Cooler	Vibration	0.968 ± 0.047	1.000 ± 0.000	+0.032
Valve	Pressure	0.683 ± 0.038	0.649 ± 0.008	-0.034
Valve	Vibration	0.867 ± 0.046	0.962 ± 0.008	+0.096
Pump	Pressure	0.908 ± 0.020	0.926 ± 0.005	+0.019
Pump	Vibration	0.924 ± 0.028	0.980 ± 0.002	+0.057
Accumulator	Pressure	0.681 ± 0.020	0.725 ± 0.013	+0.044
Accumulator	Vibration	0.871 ± 0.015	0.953 ± 0.005	+0.082

Table S81: Nearest-training-cycle audit in standardized 240-dimensional hydraulic feature space.

Task	Test cycles	Same-class nearest	Median distance	Median cosine
accumulator	213	0.258	24.760	0.829
cooler	214	1.000	2.285	0.988
pump	214	0.360	25.287	0.833
valve	213	0.465	25.246	0.834

20 Classwise calibration, selective prediction and computational scaling

All ensemble probabilities below are generated from models temperature-scaled on their calibration split. Probability averaging and logit averaging are evaluated separately. Classwise expected calibration error is the mean of six one-versus-rest, 10-bin errors. Selective risk retains observations in descending order of maximum predicted probability; it uses no target label to rank acceptance.

Table S82: Chemical fixed-ensemble probability quality under probability and logit averaging.

Method	Aggregation	Subset	AUROC	NLL	ECE	Classwise ECE
RO-CAGF	probability mean	all	0.851	1.300	0.116	0.066
RO-CAGF	probability mean	affected	0.837	1.411	0.136	0.072
RO-CAGF	logit mean	all	0.847	1.487	0.193	0.082
RO-CAGF	logit mean	affected	0.832	1.653	0.228	0.092
RO-PDRF-Full	probability mean	all	0.851	1.713	0.226	0.086
RO-PDRF-Full	probability mean	affected	0.819	2.019	0.269	0.100
RO-PDRF-Full	logit mean	all	0.848	2.032	0.274	0.100
RO-PDRF-Full	logit mean	affected	0.813	2.516	0.338	0.120

Linear-layer operations are counted with a one-observation forward pass. Two floating-point operations are assigned per multiply–accumulate. State memory assumes 32-bit parameters; forward activation memory sums linear-layer outputs and is a reproducible footprint measure, not a device-specific peak-resident-memory claim.

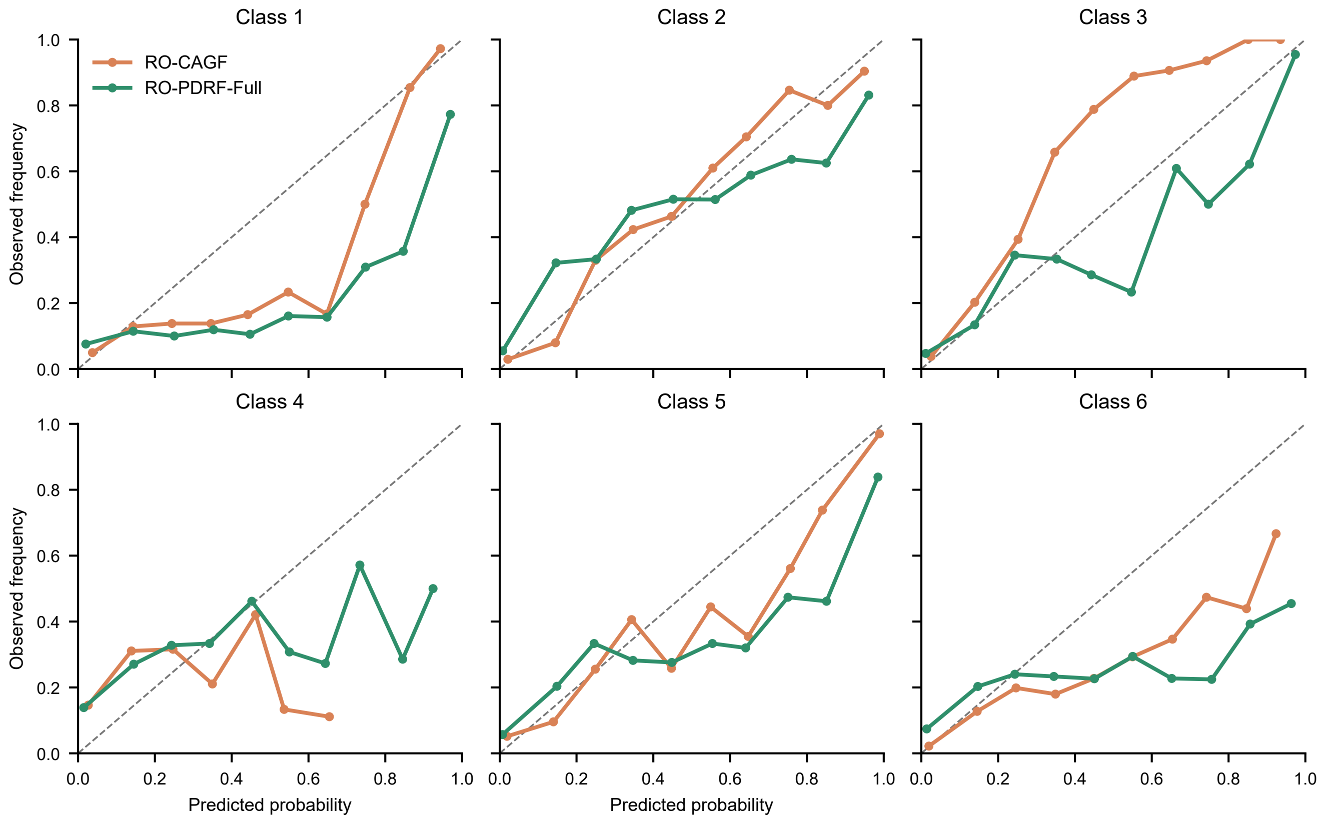

Figure S8: **Affected-row classwise calibration.** One-versus-rest predicted probability and observed class frequency for the two recovery-objective-matched ensembles. Points are shown only for non-empty bins.

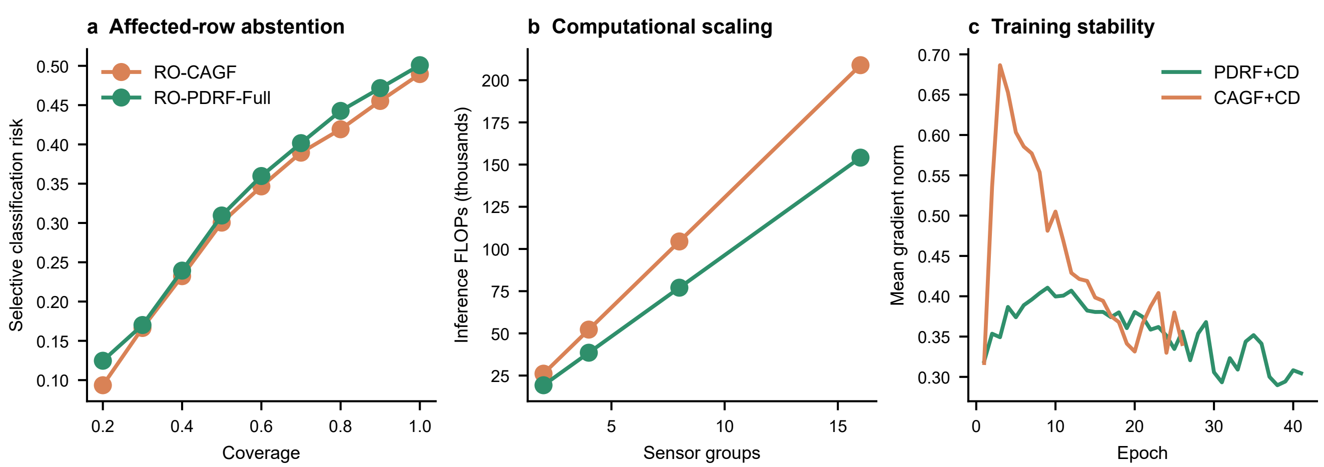

Figure S9: **Deployment and optimization audits.** **a**, affected-row selective classification risk. **b**, counted inference FLOPs versus sensor-group count. **c**, mean pre-clipping gradient norm by epoch for the two clean-distillation controls.

Table S83: Inference complexity as the number of sensor groups increases.

Method	Groups	Parameters	FLOPs	State (KiB)	Activations (KiB)
CAGF	2	13326	26112	52.1	1.1
PDRF	2	9870	19264	38.6	0.9
CAGF	4	26588	52224	103.9	1.9
PDRF	4	19740	38528	77.1	1.9
CAGF	8	53112	104448	207.5	3.5
PDRF	8	39480	77056	154.2	3.7
CAGF	16	106160	208896	414.7	6.7
PDRF	16	78960	154112	308.4	7.4

Table S84: Training stability for clean-distillation controls. Gradient norms are measured before clipping at 5.

Method	Max epochs	Median gradient norm	Maximum gradient norm	Final selection CE
CAGF+CD	26	0.458	0.766	0.587
EF-PD+CD	32	0.425	1.105	0.382
ModDrop-SD	17	0.422	1.014	0.718
PDRF+CD	41	0.373	0.540	0.457
PDRF+CD-T07	41	0.374	0.535	0.466
UF-PD+CD	46	0.212	0.353	0.456

21 Archived original-protocol analyses

The following tables and figures were generated under the original formal recovery/calibration protocol. They are retained for transparency and mechanism sensitivity, but they do not define headline method performance. Their fixed affected subset differs from the superseded seed-85000 factorial rerun; the unified headline protocol restores seed 70001 and changes the formal teacher to clean-only.

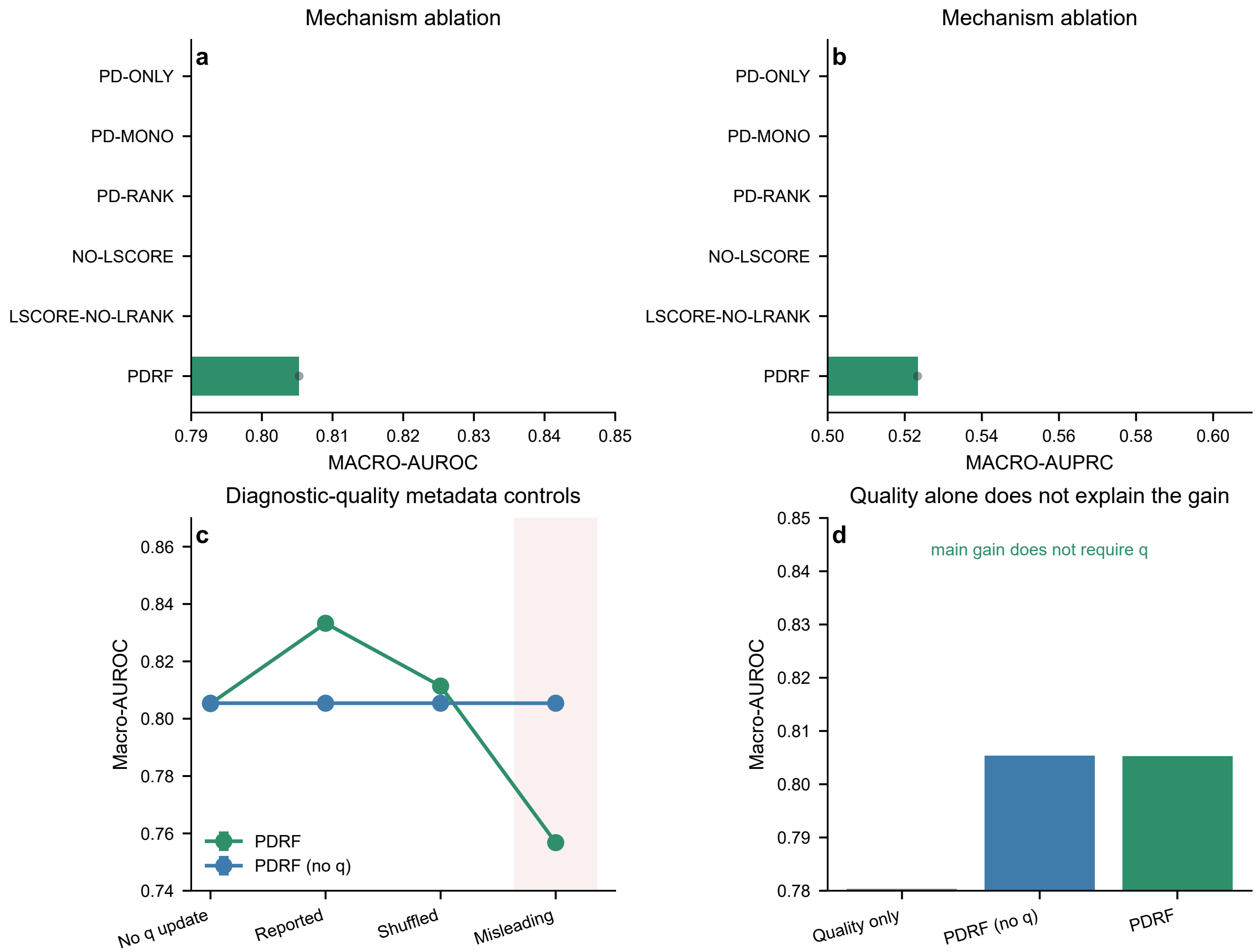

Figure S10: **Original-protocol mechanism and diagnostic-quality controls.** Mechanism ablations, metadata perturbations and a quality-only comparator are retained as sensitivity evidence.

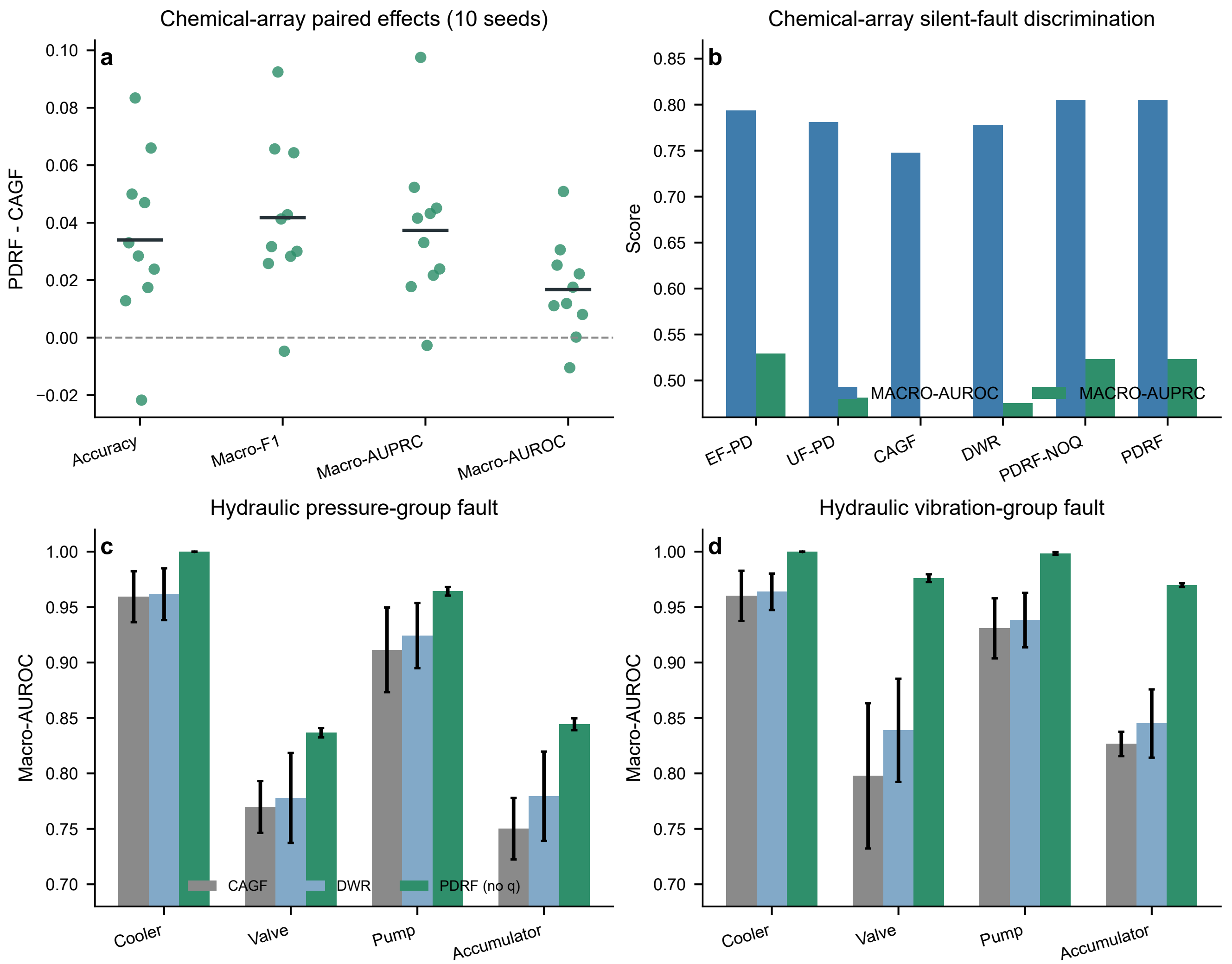

Figure S11: **Original-protocol controlled-fault comparisons.** Chemical seed-level effects and hydraulic task results are retained for protocol transparency; the clean-teacher recovery-objective-matched figure supplies the headline comparison.

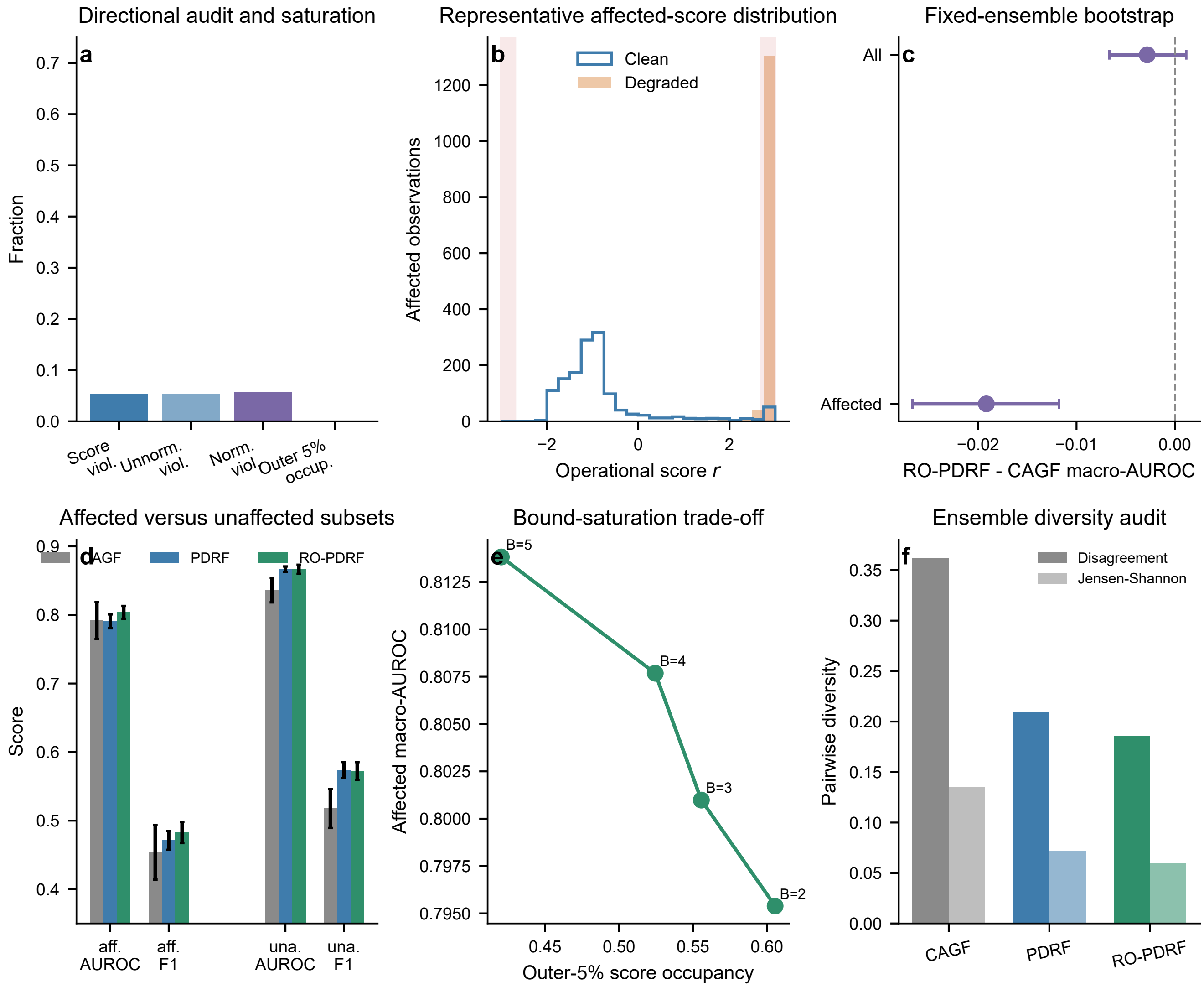

Figure S12: **Boundary analyses and recovery audits.** Directional violations, bounded-response saturation, fixed-ensemble bootstrap effects, affected/unaffected outcomes, bound sensitivity and ensemble diversity. This dense diagnostic figure is moved from the main paper to preserve the complete evidence without overloading the principal narrative.

Table S85: Original formal-run comparator results under the fixed chemical-array silent fault.

Fault	Method	Accuracy	Macro-F1	Macro-AUPRC	Macro-AUROC	NLL
natural	EF_PD	0.585	0.579	0.632	0.851	1.730
natural	UF_PD	0.575	0.561	0.612	0.856	1.650
natural	CAGF	0.535	0.509	0.569	0.834	1.670
natural	DWR	0.536	0.511	0.570	0.834	1.703
natural	PDRF_NOQ	0.590	0.566	0.642	0.866	1.793
natural	PDRF	0.594	0.568	0.644	0.867	1.849
silent	EF_PD	0.524	0.518	0.549	0.805	2.418
silent	UF_PD	0.527	0.518	0.549	0.814	2.251
silent	CAGF	0.516	0.493	0.542	0.817	1.828
silent	DWR	0.515	0.494	0.538	0.815	1.899
silent	PDRF_NOQ	0.548	0.533	0.583	0.835	2.233
silent	PDRF	0.550	0.535	0.580	0.834	2.345

Table S86: Original formal-run mechanism ablation.

Variant	Accuracy	Macro-AUPRC	Macro-AUROC	NLL
PD_ONLY	0.531	0.556	0.817	2.138
PD_MONO	0.537	0.563	0.821	2.071
PD_RANK	0.514	0.547	0.812	2.309
NO_LSCORE	0.514	0.545	0.811	2.312
LSCORE_NO_LRANK	0.547	0.555	0.822	2.441
PDRF	0.550	0.580	0.834	2.345

Table S87: Original formal-run recovery and calibration results. Aff. denotes the affected subset.

Method	All Acc.	All AUROC	Aff. Acc.	Aff. F1	Aff. AUPRC	Aff. AUROC	Aff. NLL	Aff. ECE
CAGF	0.516	0.817	0.474	0.454	0.505	0.792	2.095	0.277
PDRF	0.550	0.834	0.476	0.471	0.509	0.791	3.085	0.369
RO-PDRF-Full	0.558	0.840	0.493	0.483	0.527	0.804	2.693	0.344

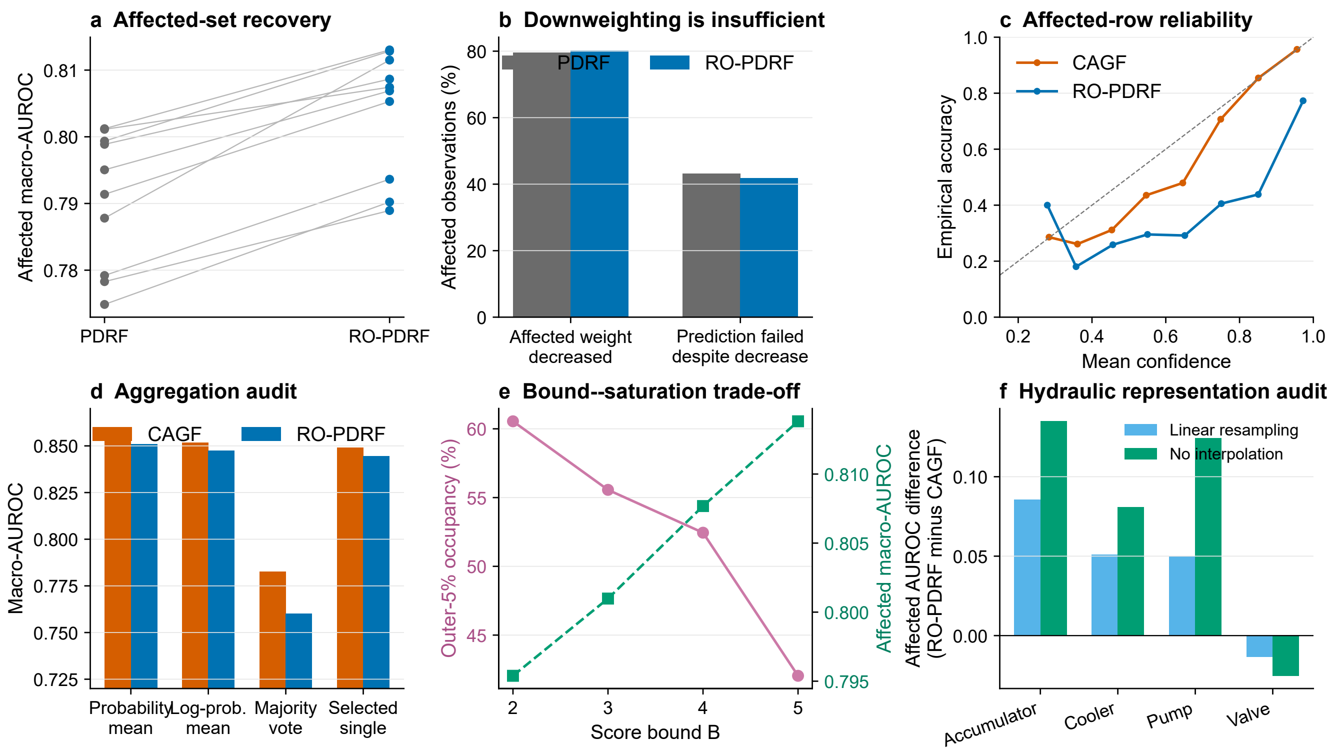

Figure S13: **Prediction-level recovery and deployment audits.** Matched recovery, failures despite downweighting, affected-row calibration, aggregation rules, score-bound sensitivity and hydraulic representation robustness.

22 External public-data feasibility audit

We searched for an external dataset that simultaneously provides (i) naturally occurring or maintenance-confirmed sensor failure, (ii) an independent downstream state to recover after suppressing the faulty sensor and (iii) multiple physical devices or sites. The NASA sensor-fault resource (`https://data.nasa.gov/dataset/modeling-detection-and-disambiguation-of-sensor-faults-for-aerospace-applications`) describes experimental actuator data and simulated sensor faults, but its public landing page does not provide a directly usable natural sensor-failure cohort with an independent recovery target. BASiC (`https://doi.org/10.5281/zenodo.8195068`) contains 70 autonomous aerial-vehicle flights, but its failures are introduced through software-in-the-loop rather than documented natural hardware sensor failures. ALFA (`https://theairlab.org/alfa-dataset/`) contains real flight records with engine and control-surface failures, which are actuator or system failures rather than the required sensor-failure annotation. The AHU data used here contain operational sensor-fault labels but no independent downstream recovery target; the hydraulic data contain native component targets but controlled sensor interventions and one rig. We therefore did not relabel actuator failures as sensor faults or construct a circular downstream target. The absence of a qualifying public dataset remains a limitation rather than evidence of external field recovery.

23 Reproducibility and scope

The Q1 entry points are `run_cee_lite_routing_validation.py`, `analyse_cee_lite_results.py`, `analyse_cee_q1_scores.py`, `analyse_cee_fault_plausibility.py` and `make_cee_q1_figures.py`. The machine-readable frozen design is `cee_cf10_r2_lite_config.json`. Exact local package versions are recorded in `requirements-lock.txt` and `environment.yml`. Prediction-level outputs, selector thresholds, safety accounting, model costs, probability metrics and fixed-mechanism/batch intervals are stored under `source_data/q1_*`. Figures use Python/matplotlib exclusively. Editable SVG/PDF figures and 600-dpi TIFFs are included. The versioned release archive contains a SHA-256 manifest, and the public repository URL and exact commit are reported in the main article.

The chemical-array experiment uses real longitudinal measurements with controlled feature-level sensor faults. The hydraulic experiment uses native test-rig component-condition labels with controlled sensor faults. The AHU audit adds naturally observed operational records with expert-annotated sensor-fault classes from 41 installed AHUs across three buildings. Those labels were assigned from engineering rules and expert review rather than maintenance-confirmed sensor replacement records. The evidence supports a controlled-corruption risk-control protocol and a field-operational direct-diagnosis boundary, not hardware-failure prognosis, downstream field recovery or zero-shot building transfer.